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Ahn et al.

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(54) **SEMICONDUCTOR DEVICE AND DATA STORAGE SYSTEM INCLUDING THE SAME**

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(57)

ABSTRACT

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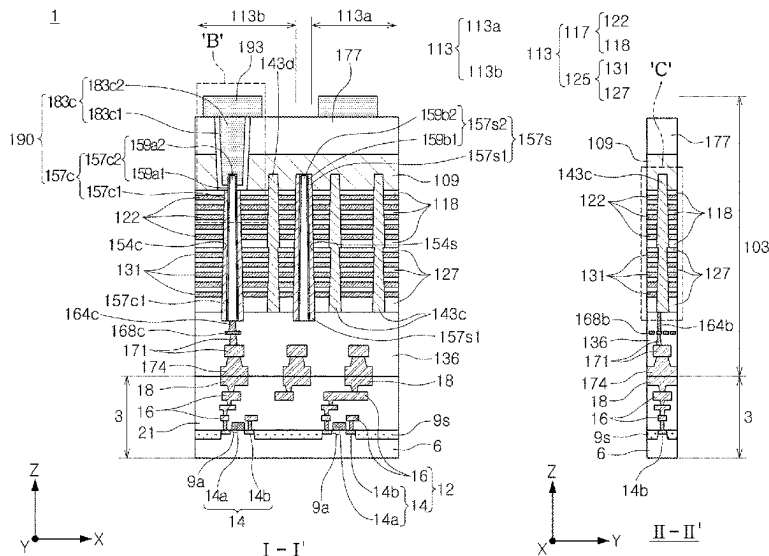
(58) **Field of Classification Search**

None

See application file for complete search history.

A semiconductor device and a data storage system including the same, the semiconductor device including: a first structure including a peripheral circuit; and a second structure, including: a pattern structure; an upper insulating layer; a stack structure between the first structure and the pattern structure and including first and second stack portions spaced apart from each other, the first and second stack portions respectively including horizontal conductive layers and interlayer insulating layers alternately stacked; separation structures penetrating through the stack structure; memory vertical structures penetrating through the first stack portion; and a contact structure penetrating through the second stack portion, the pattern structure, and the upper insulating layer, wherein the contact structure includes a lower contact plug penetrating through at least the second stack portion and an upper contact plug contacting the lower contact plug and extending upwardly to penetrate through the pattern structure and the upper insulating layer.

20 Claims, 30 Drawing Sheets



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H10B 43/35 (2023.01)
H10B 43/40 (2023.01)

(52) **U.S. Cl.**

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43/10 (2023.02); **H10B 43/27** (2023.02);
H10B 43/35 (2023.02); **H10B 43/40** (2023.02)

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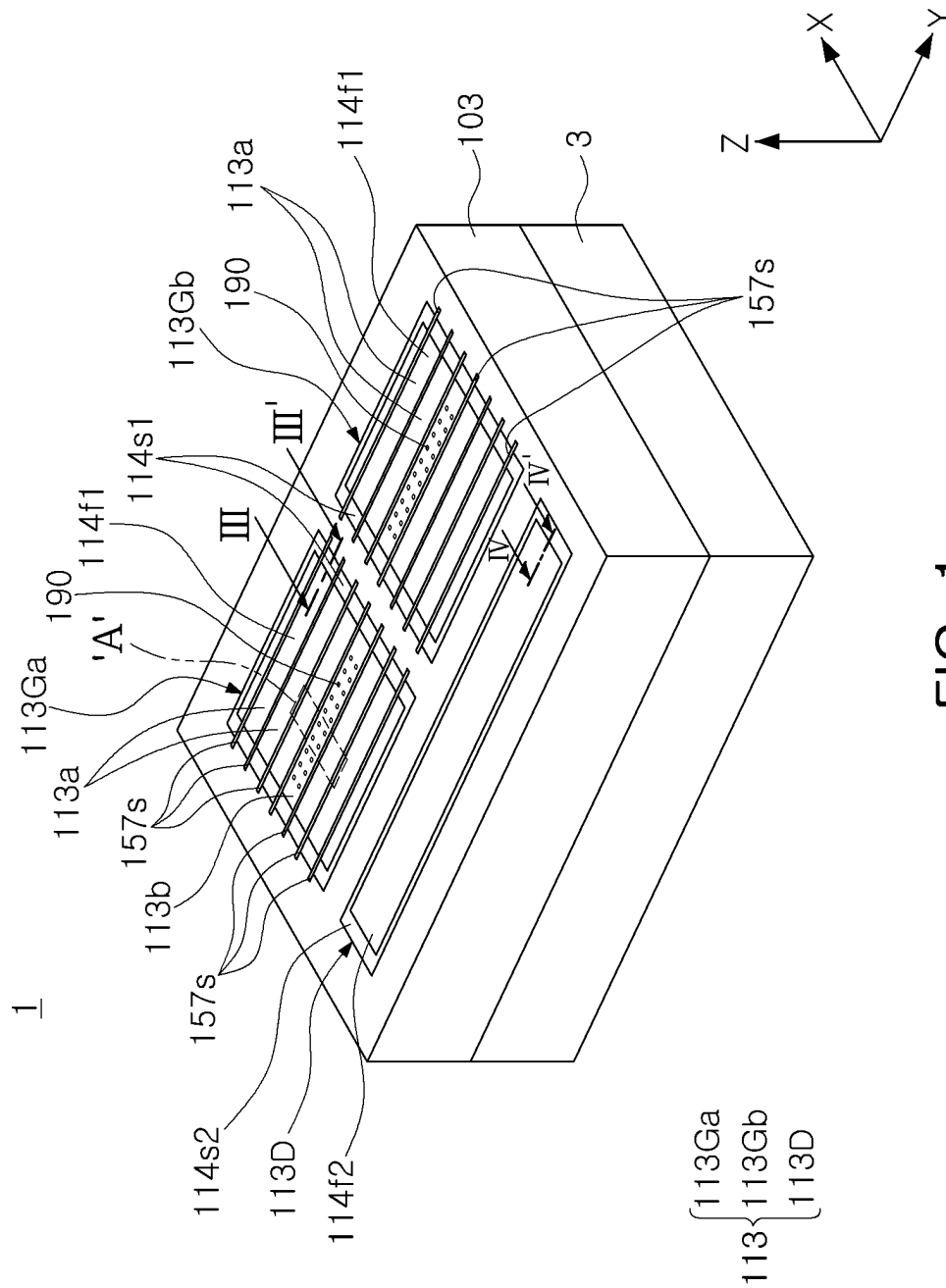
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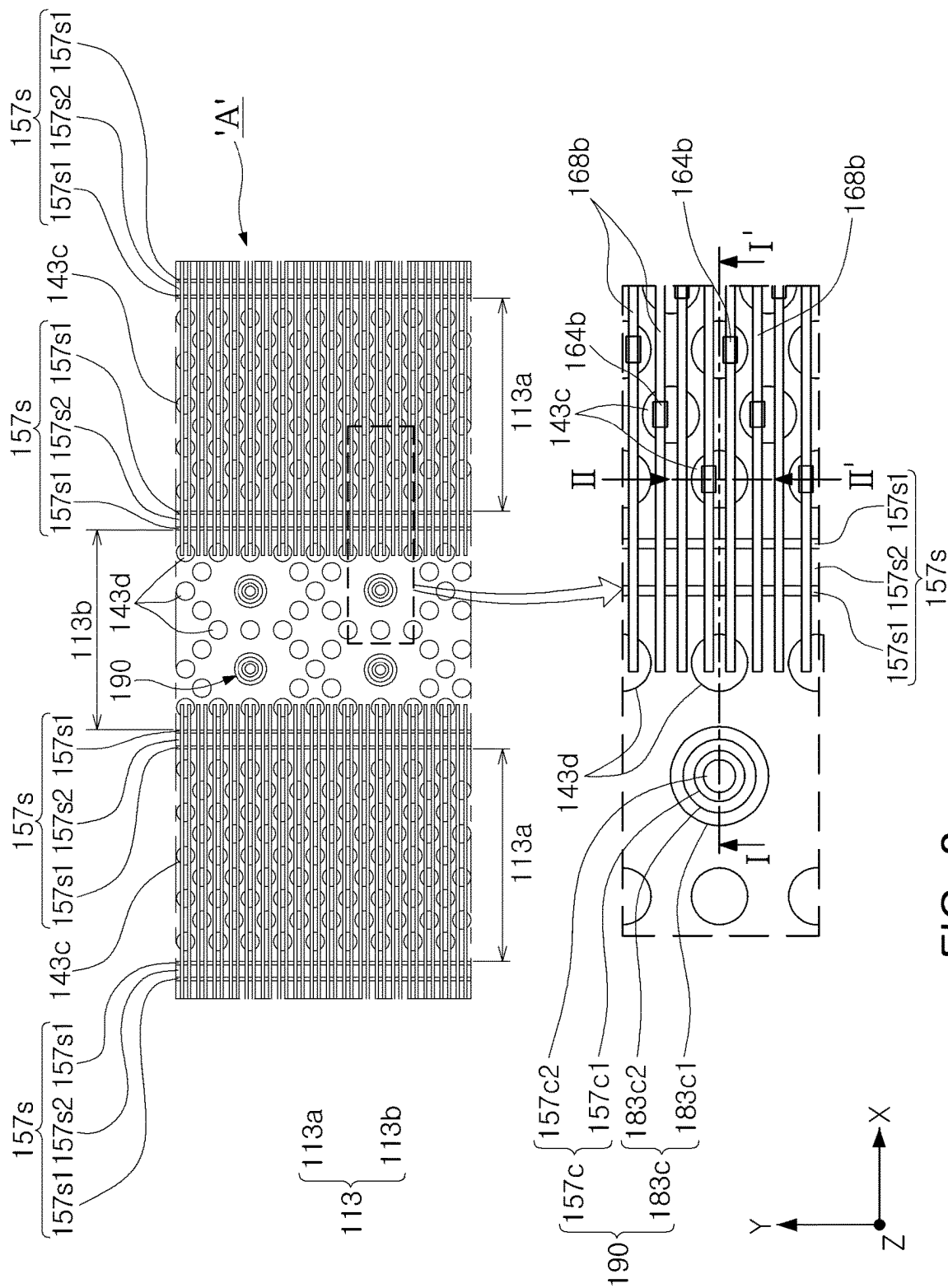
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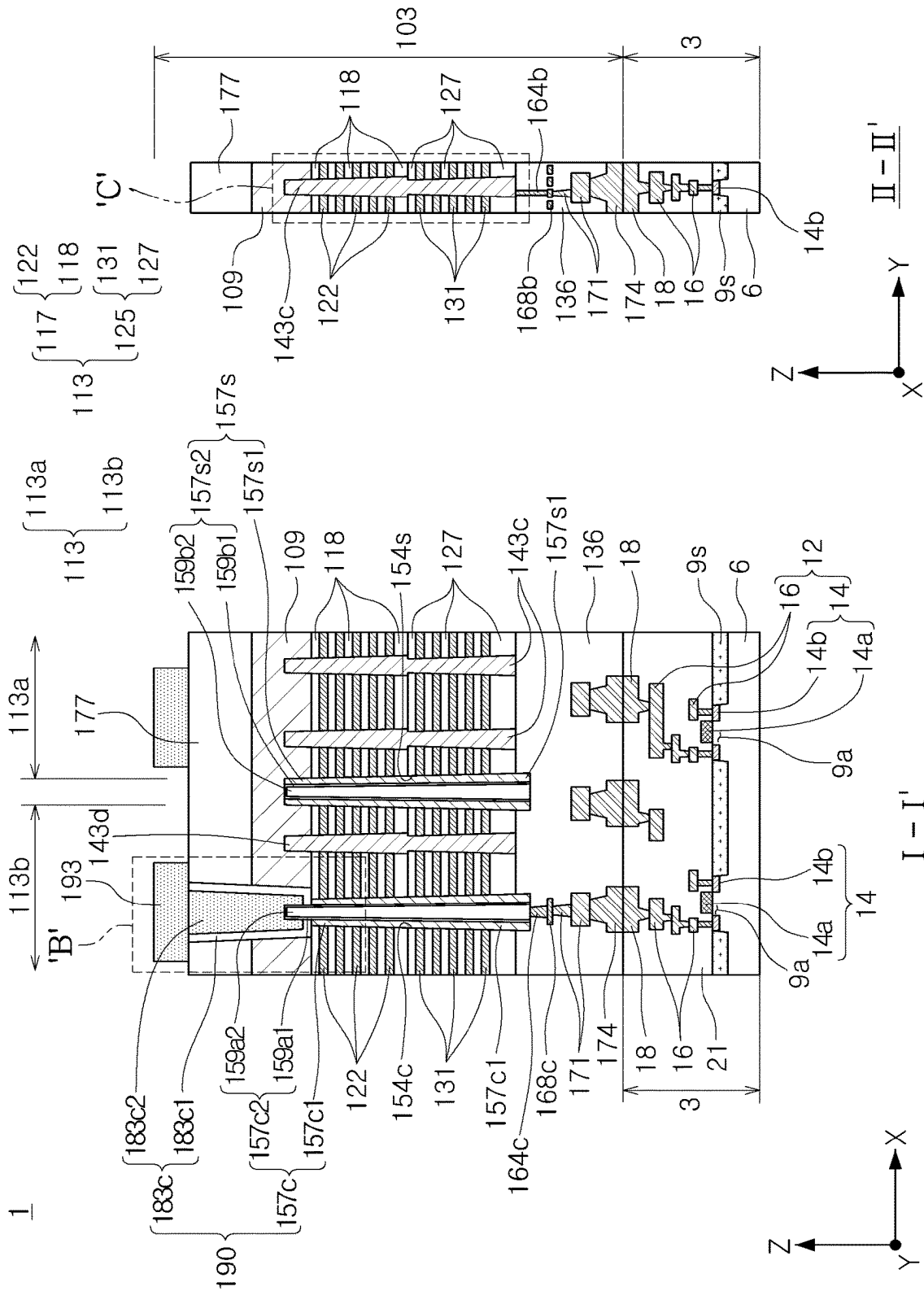
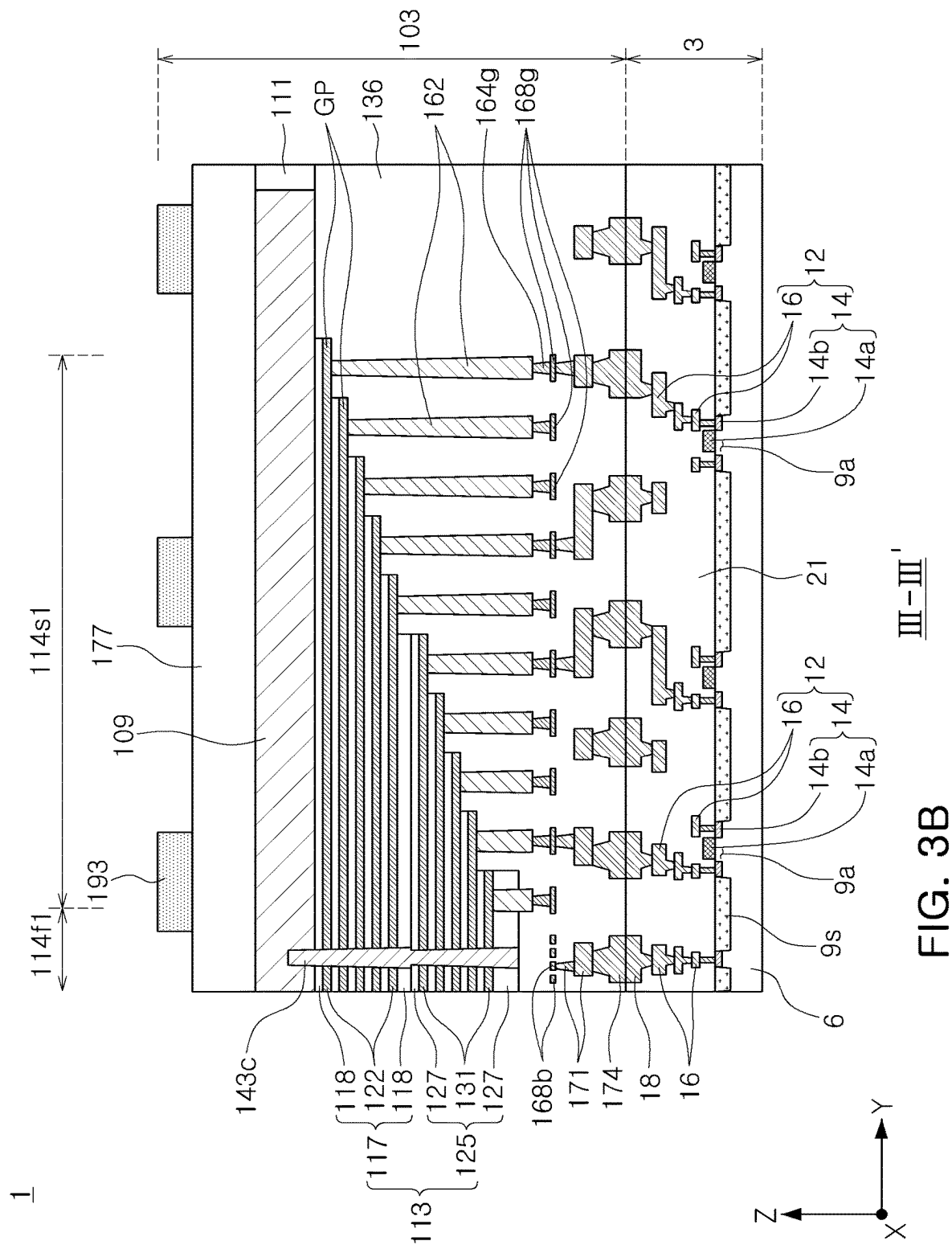


FIG. 3A



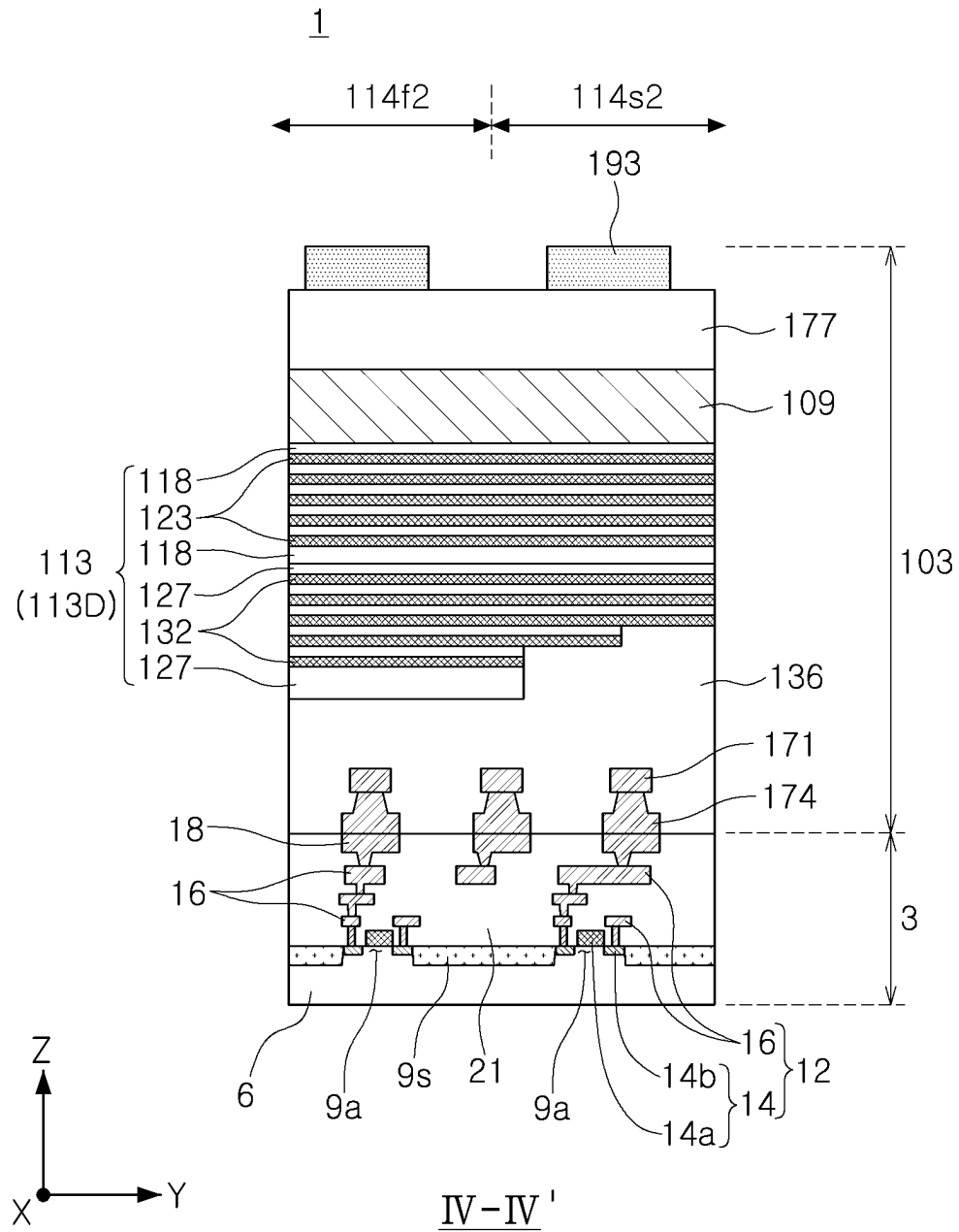


FIG. 3C

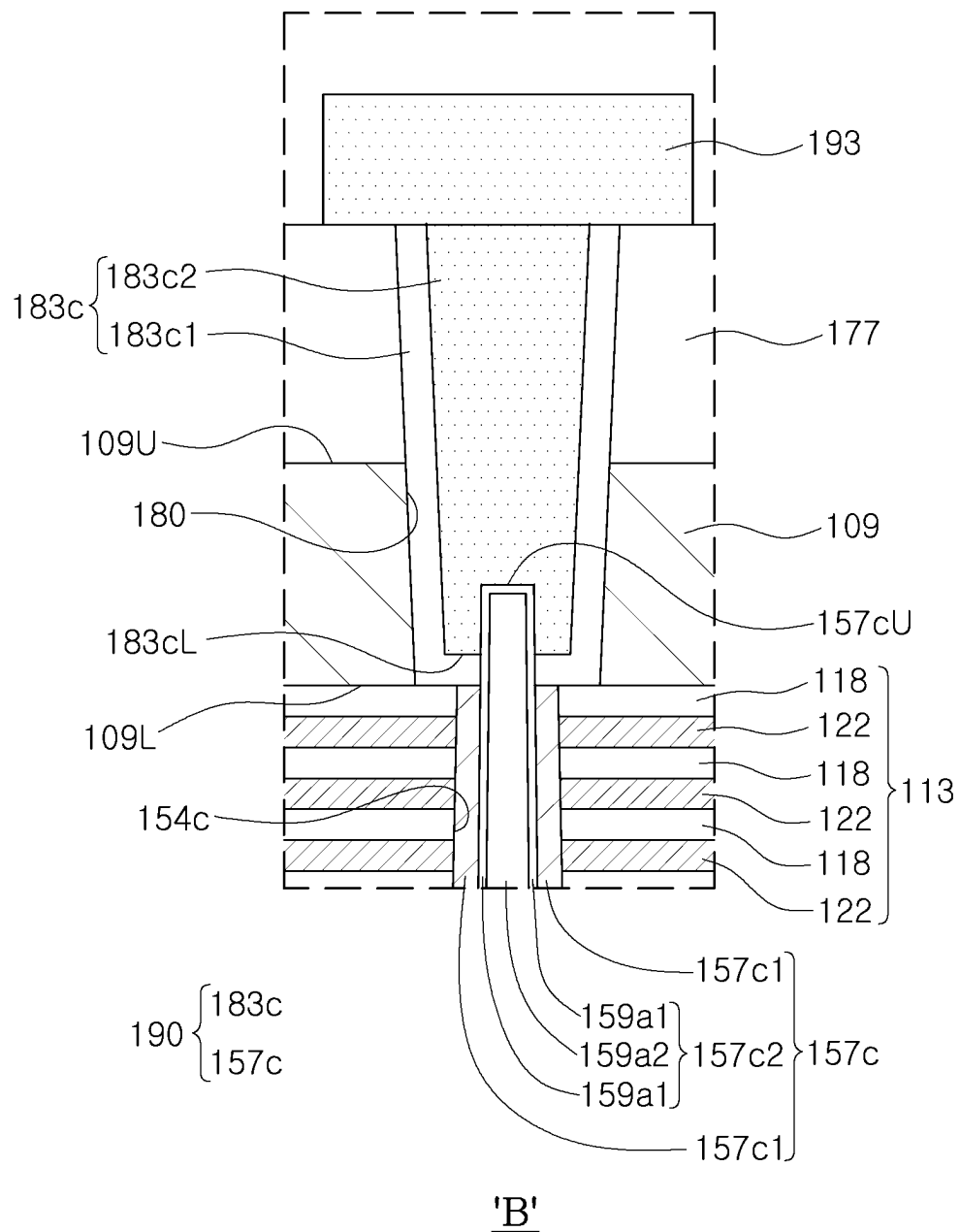


FIG. 3D

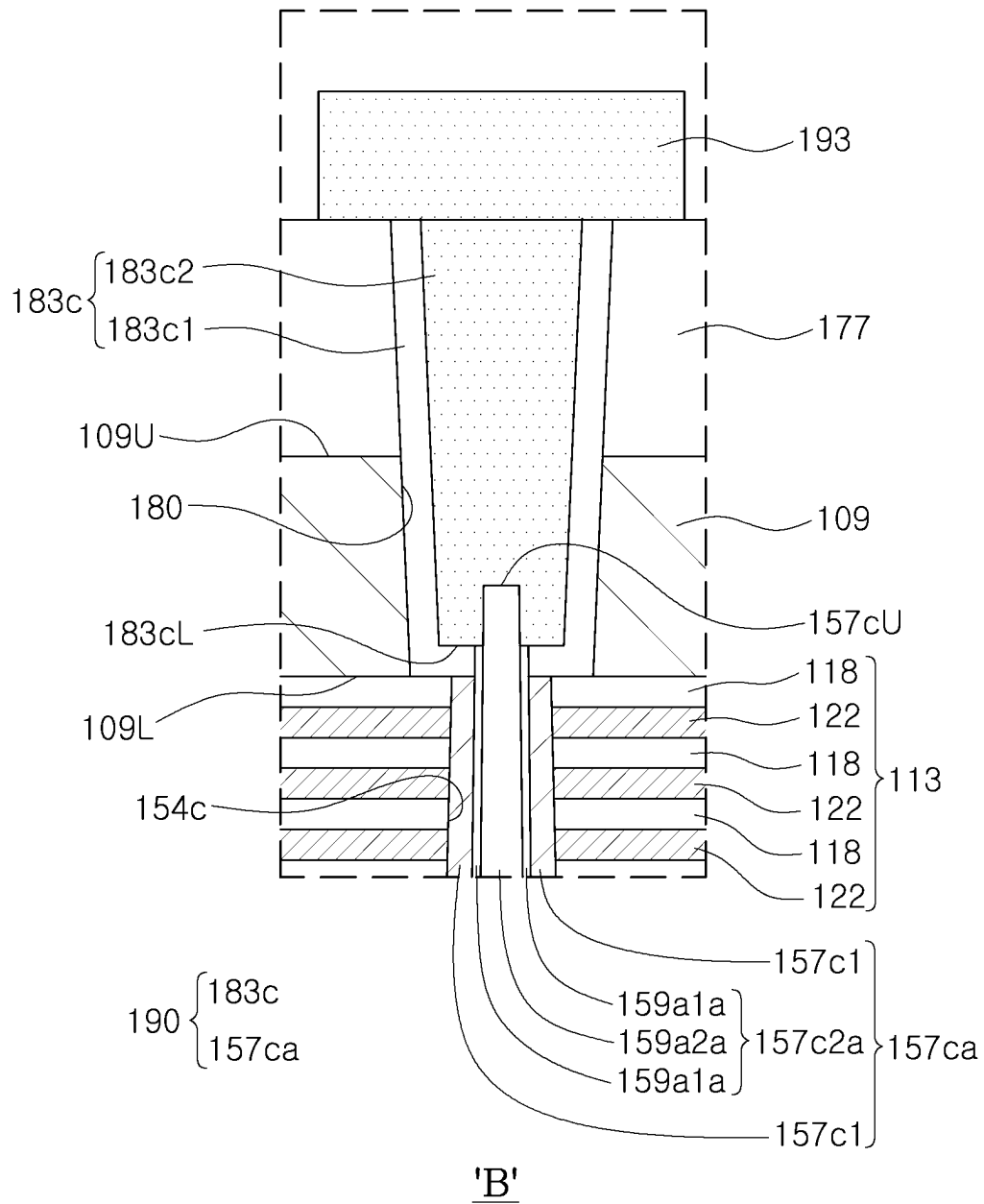


FIG. 4A

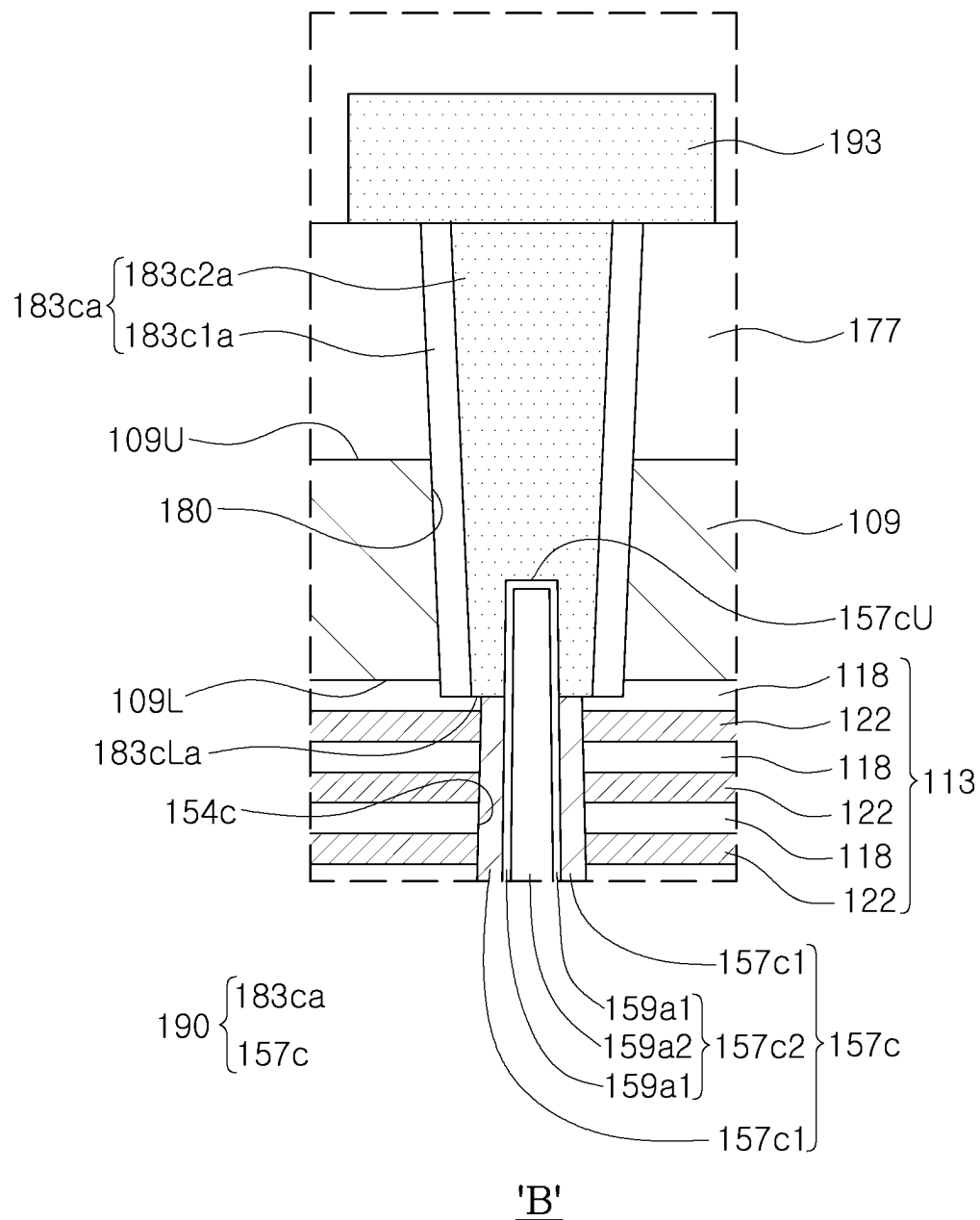


FIG. 4B

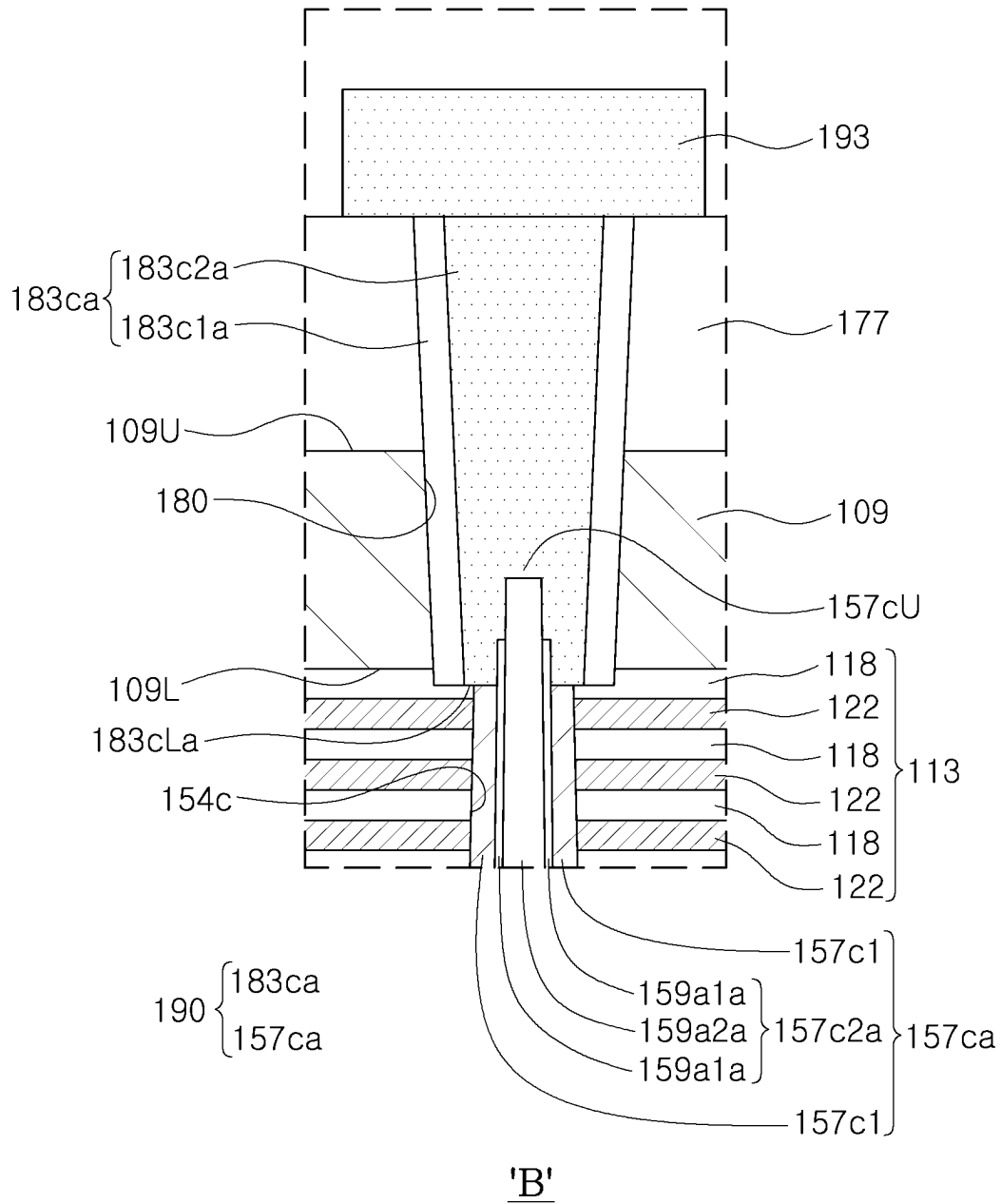


FIG. 4C

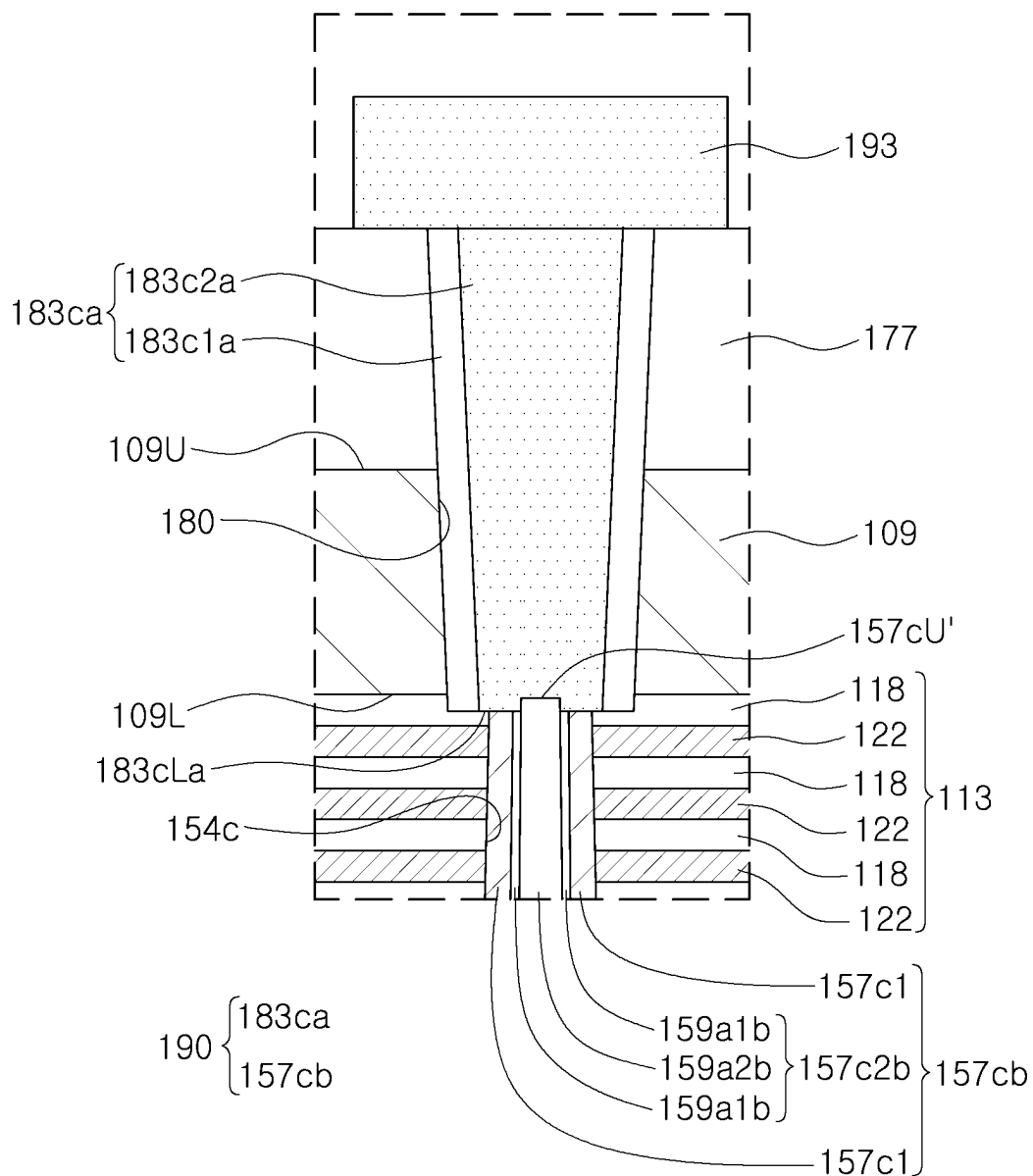


FIG. 4D

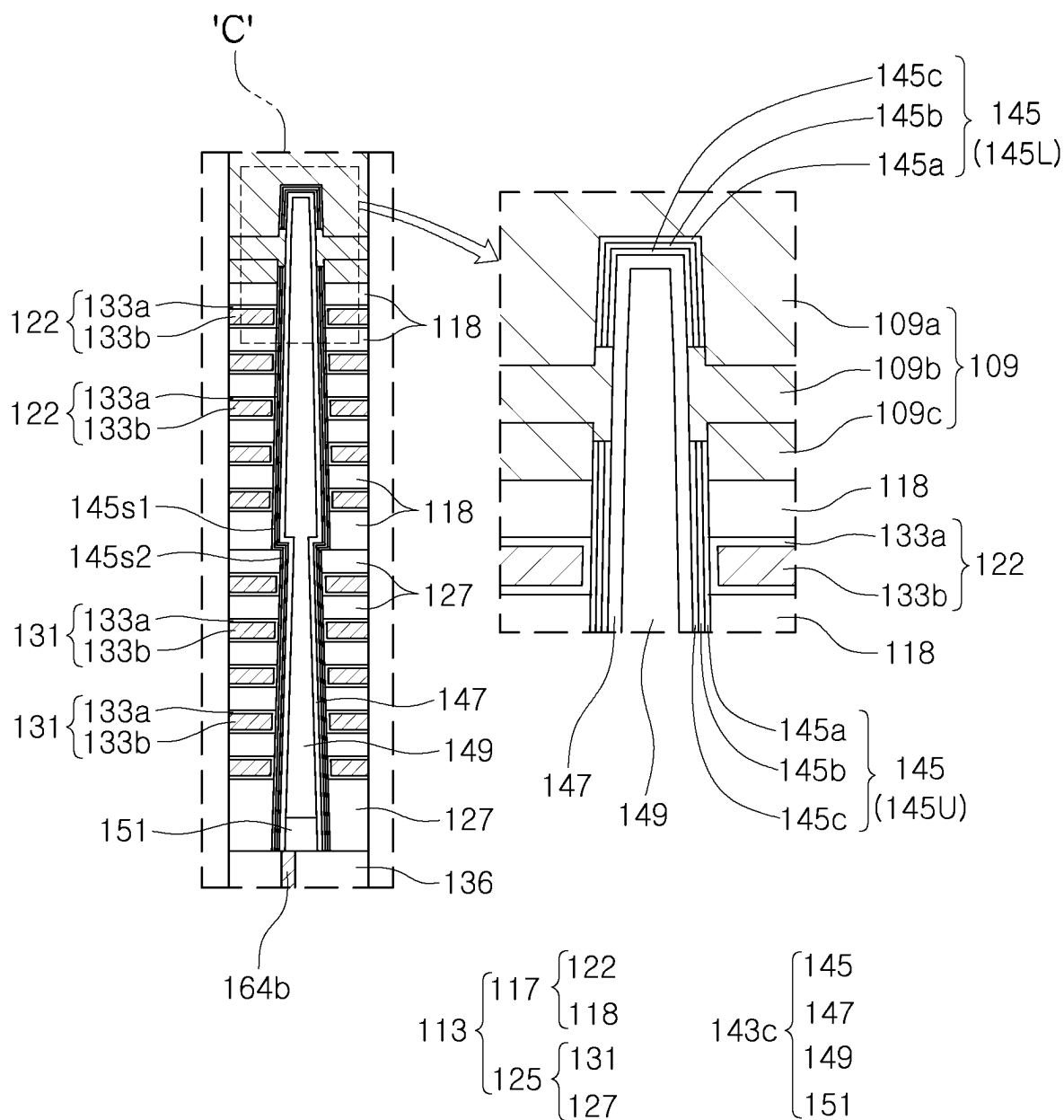


FIG. 5A

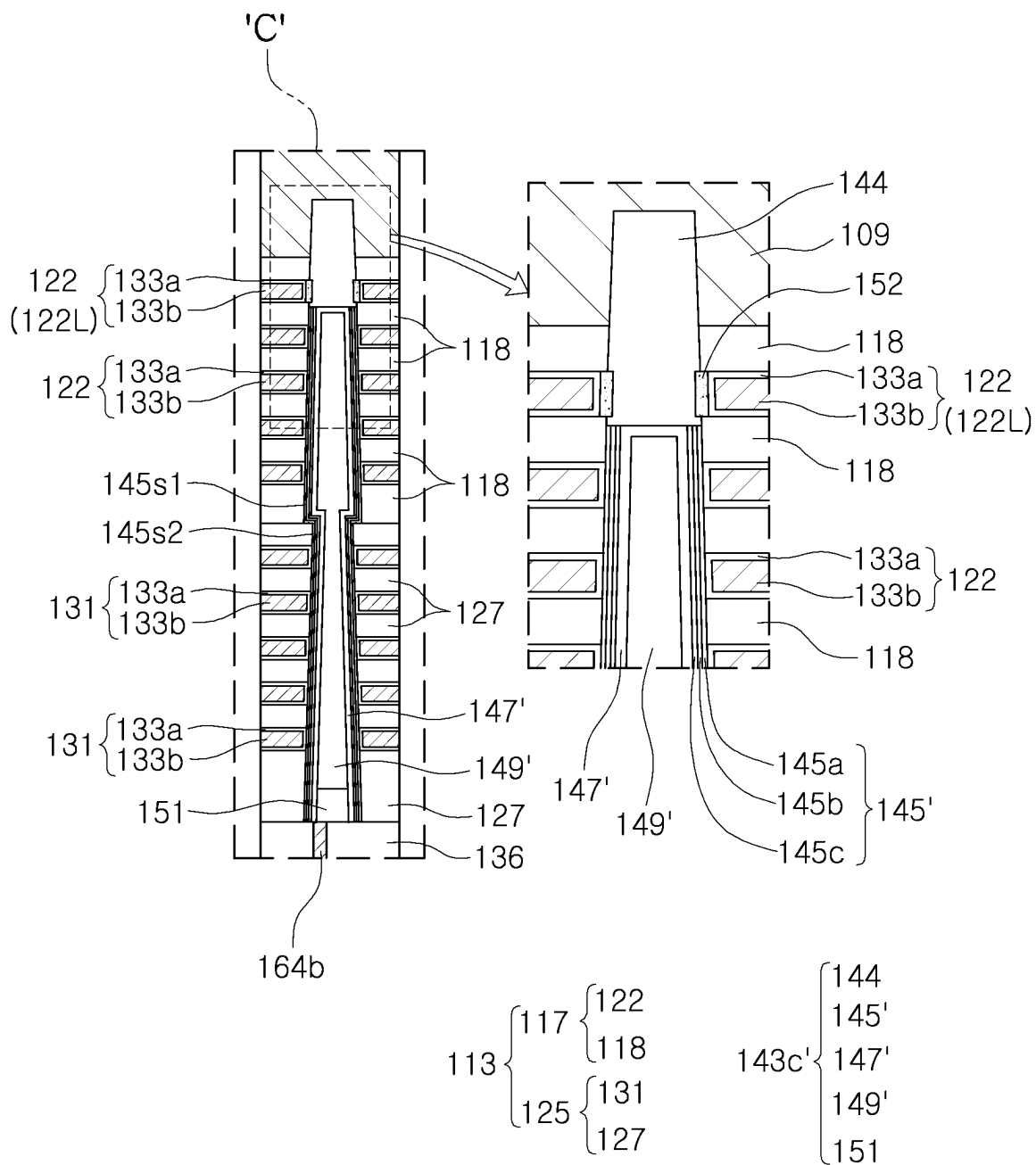


FIG. 5B

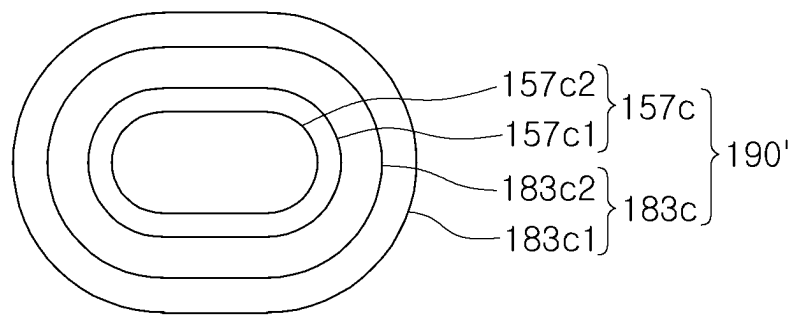


FIG. 6

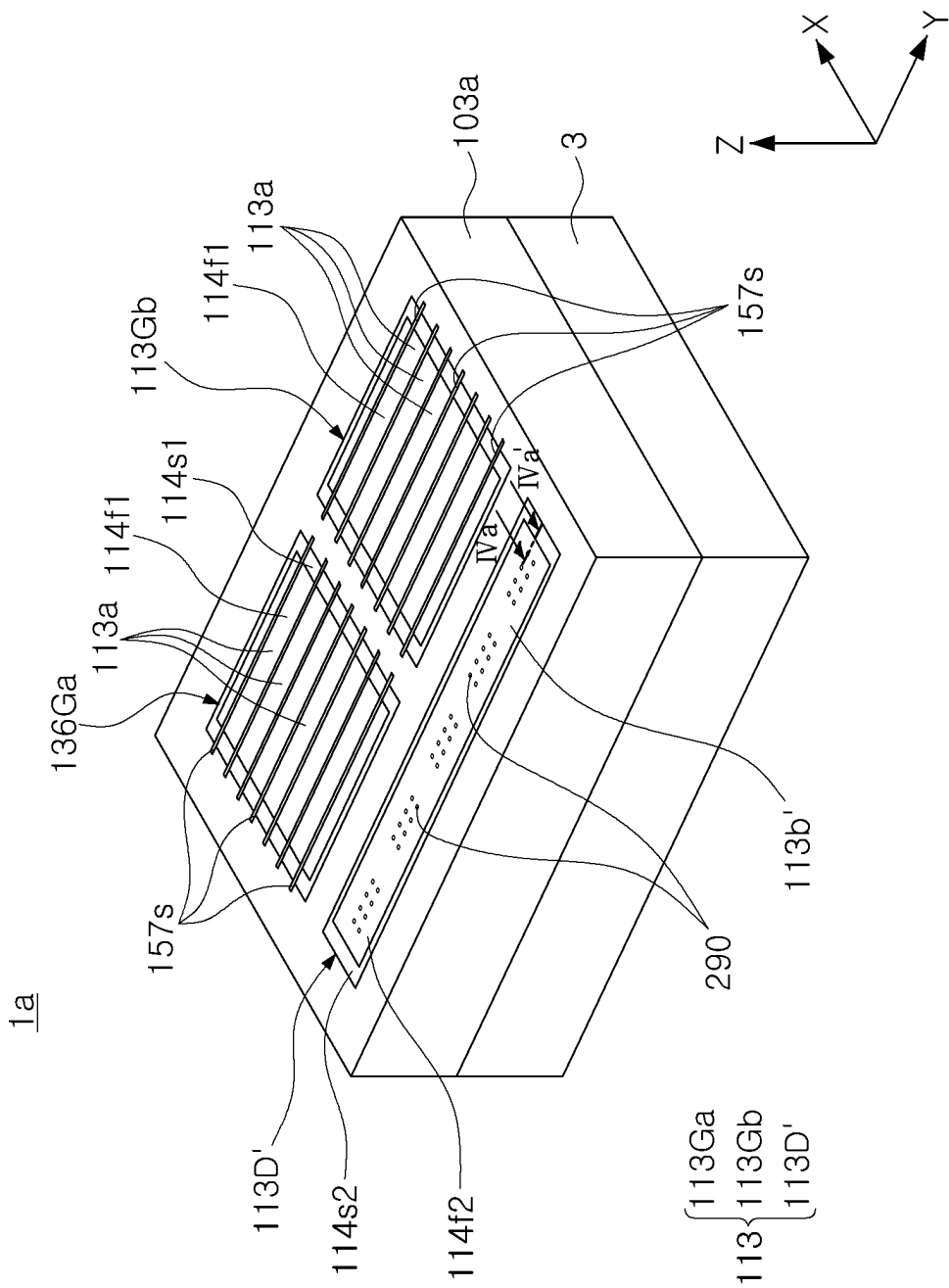
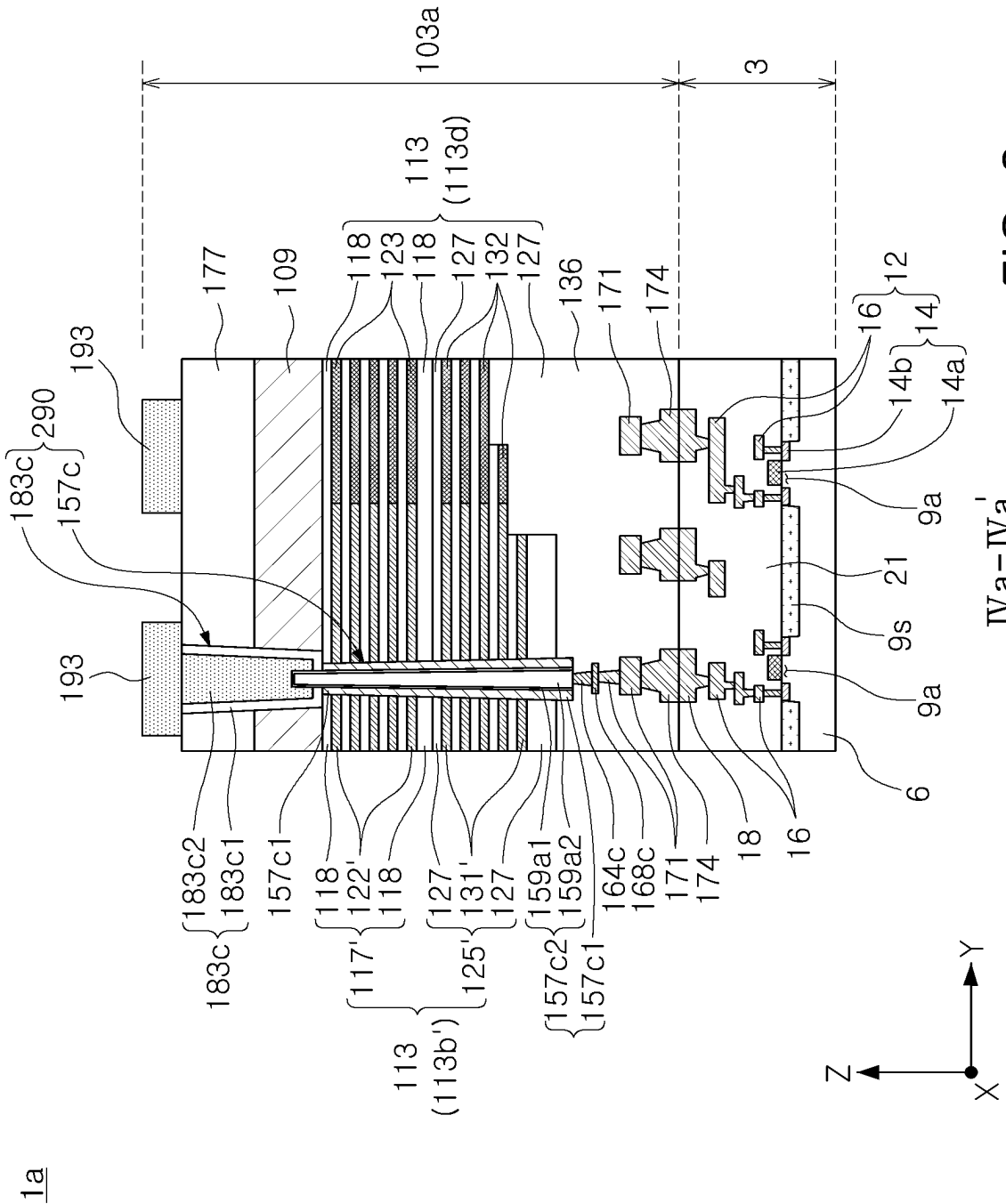


FIG. 7



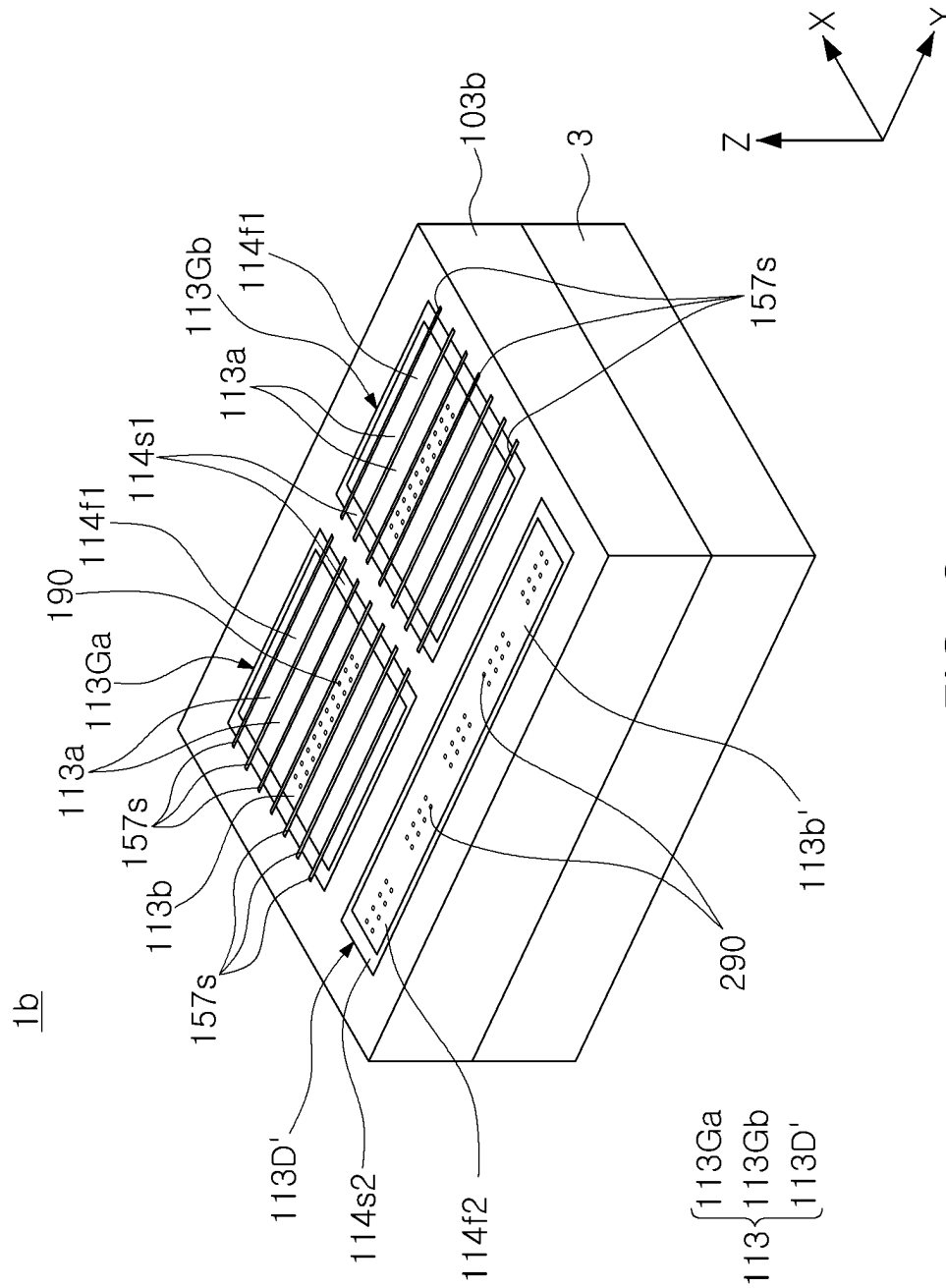


FIG. 9

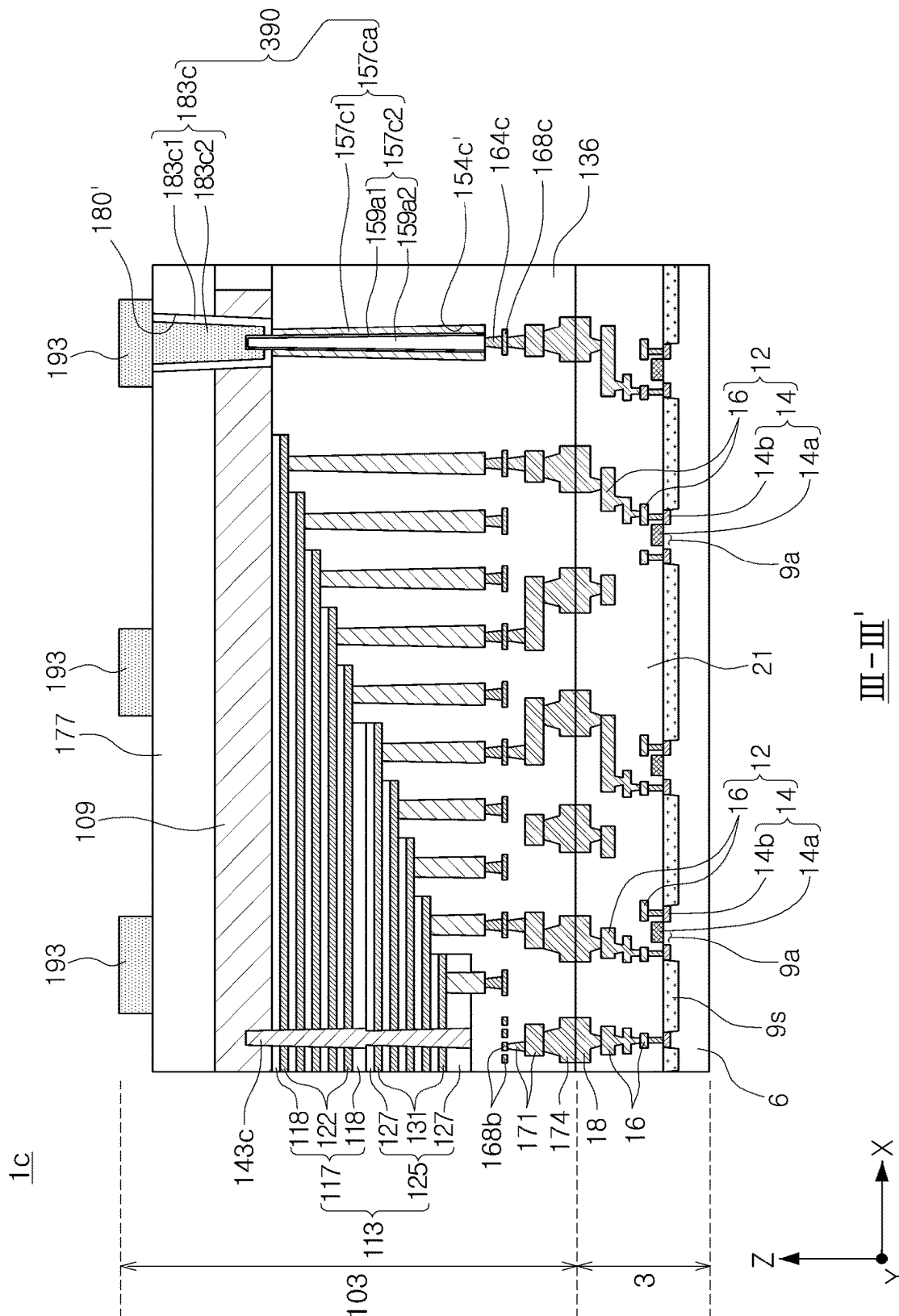
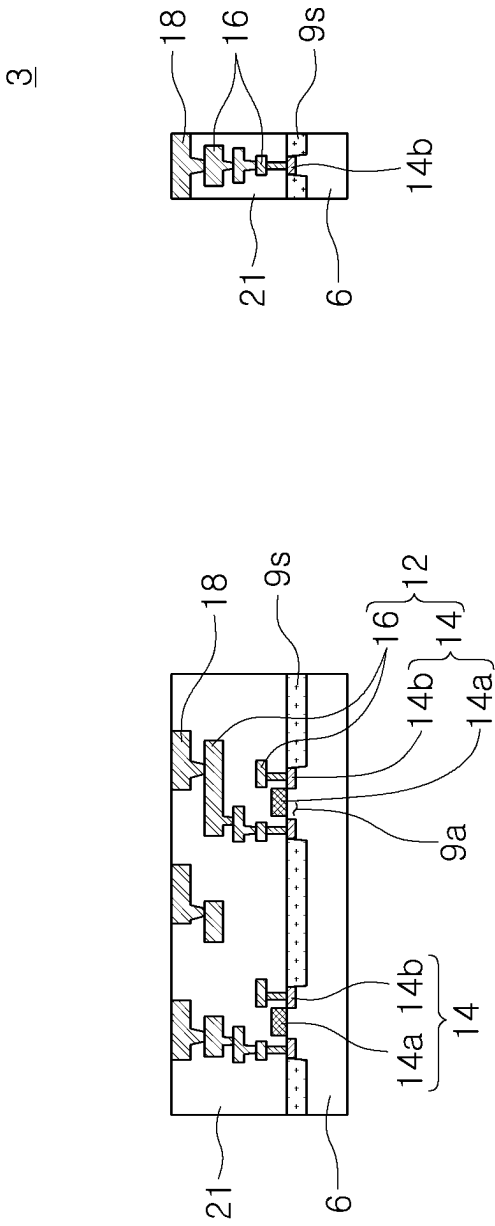


FIG. 10



I - I'

II - II'

FIG. 11

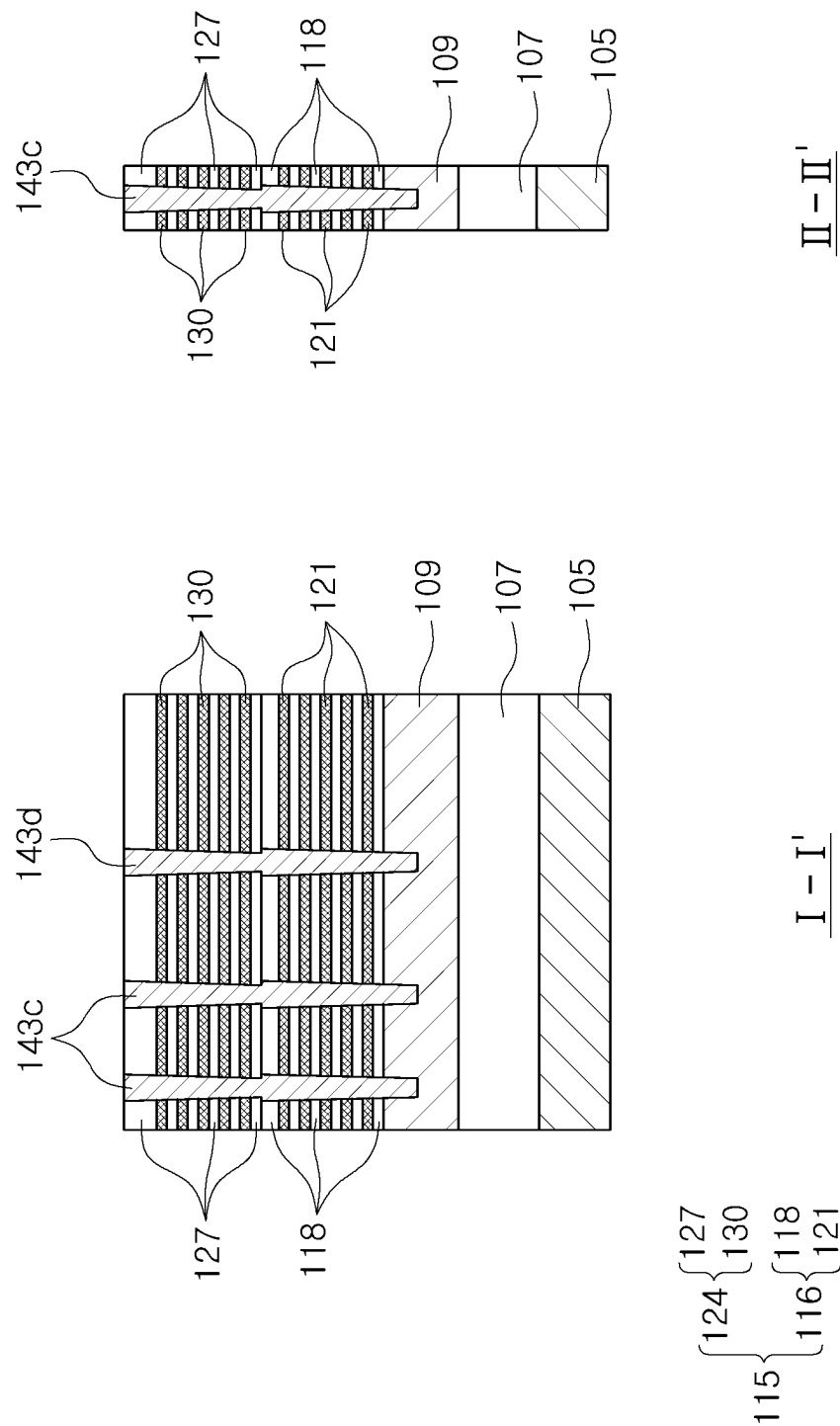
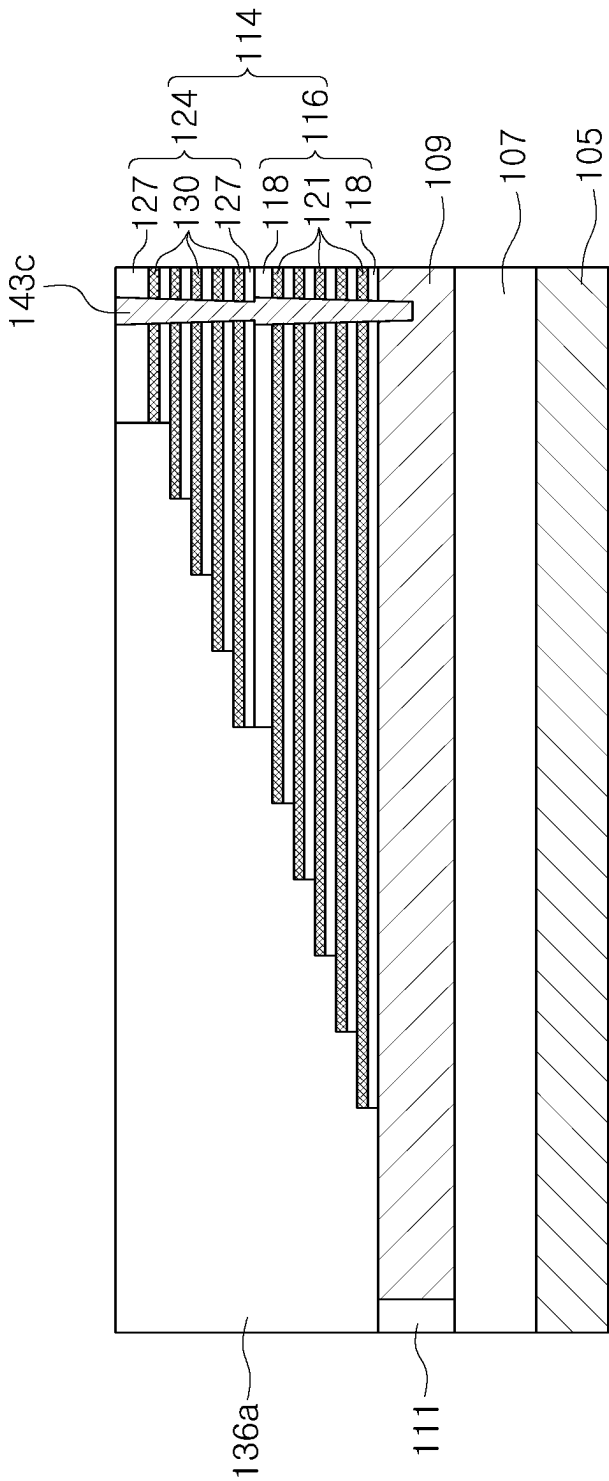
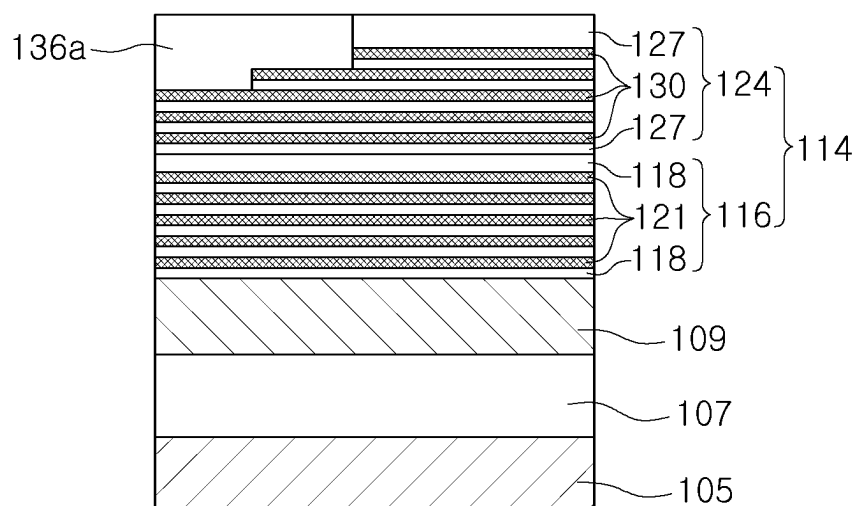


FIG. 12A



III-III'

FIG. 12B



IVa-IVa'

FIG. 12C

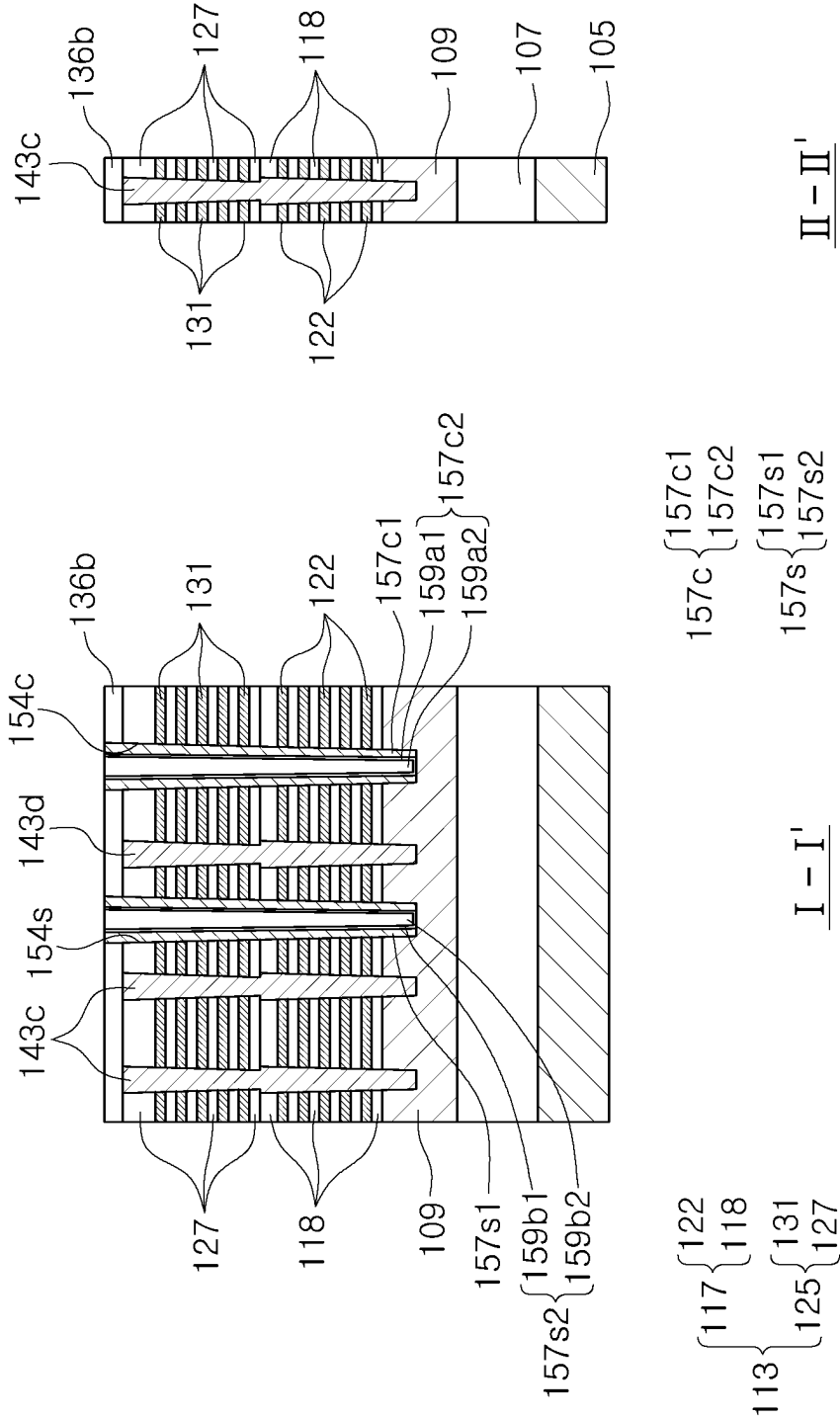
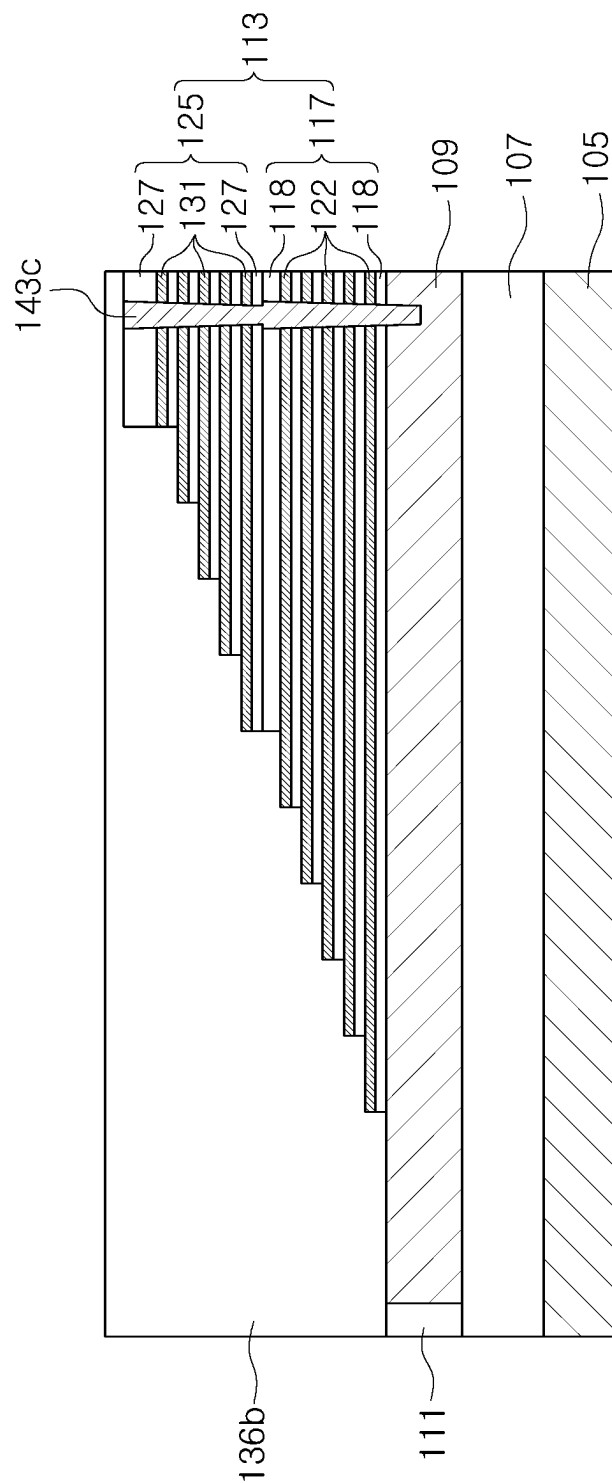
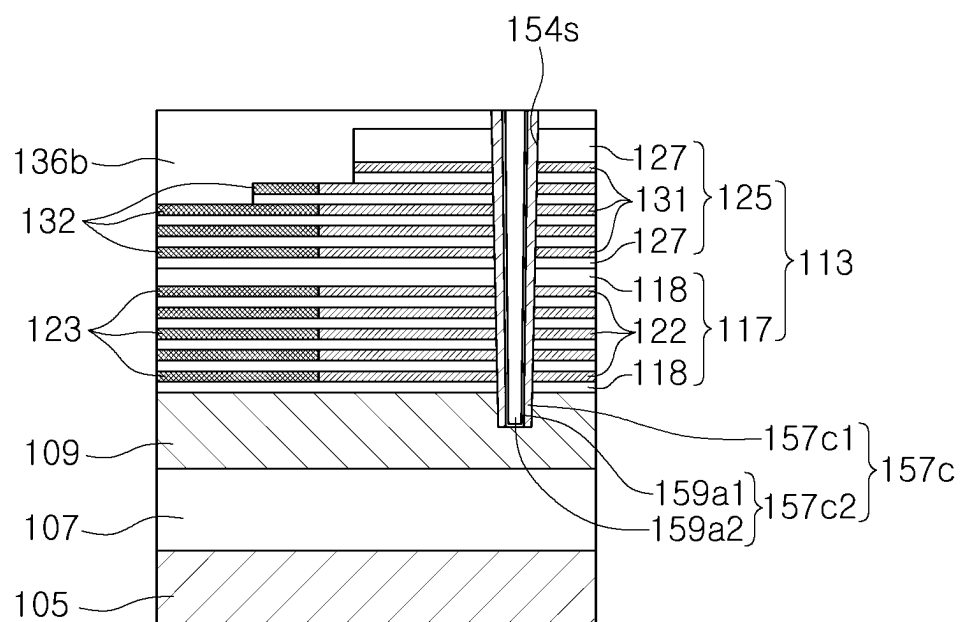


FIG. 13A



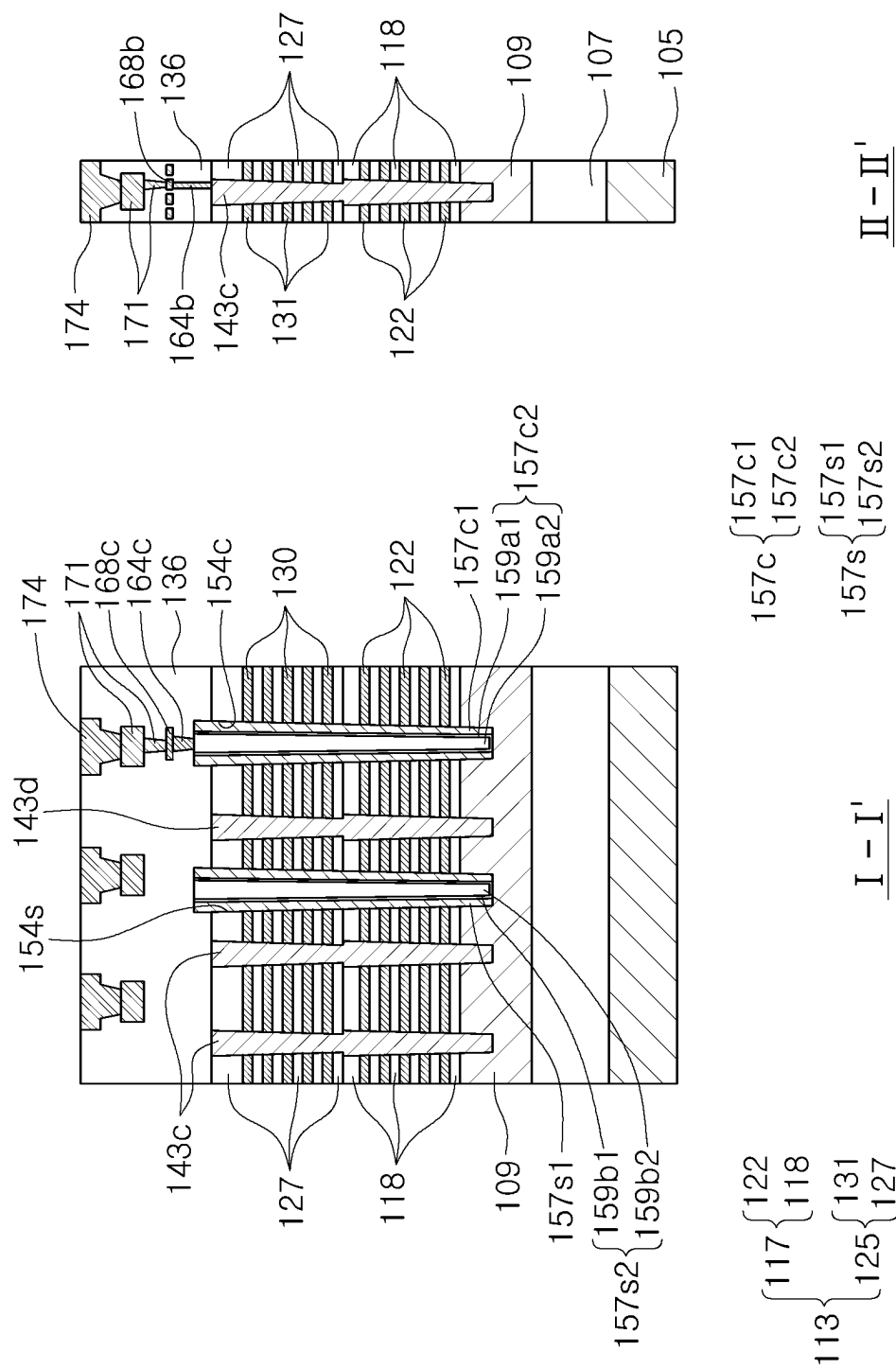
III-III'

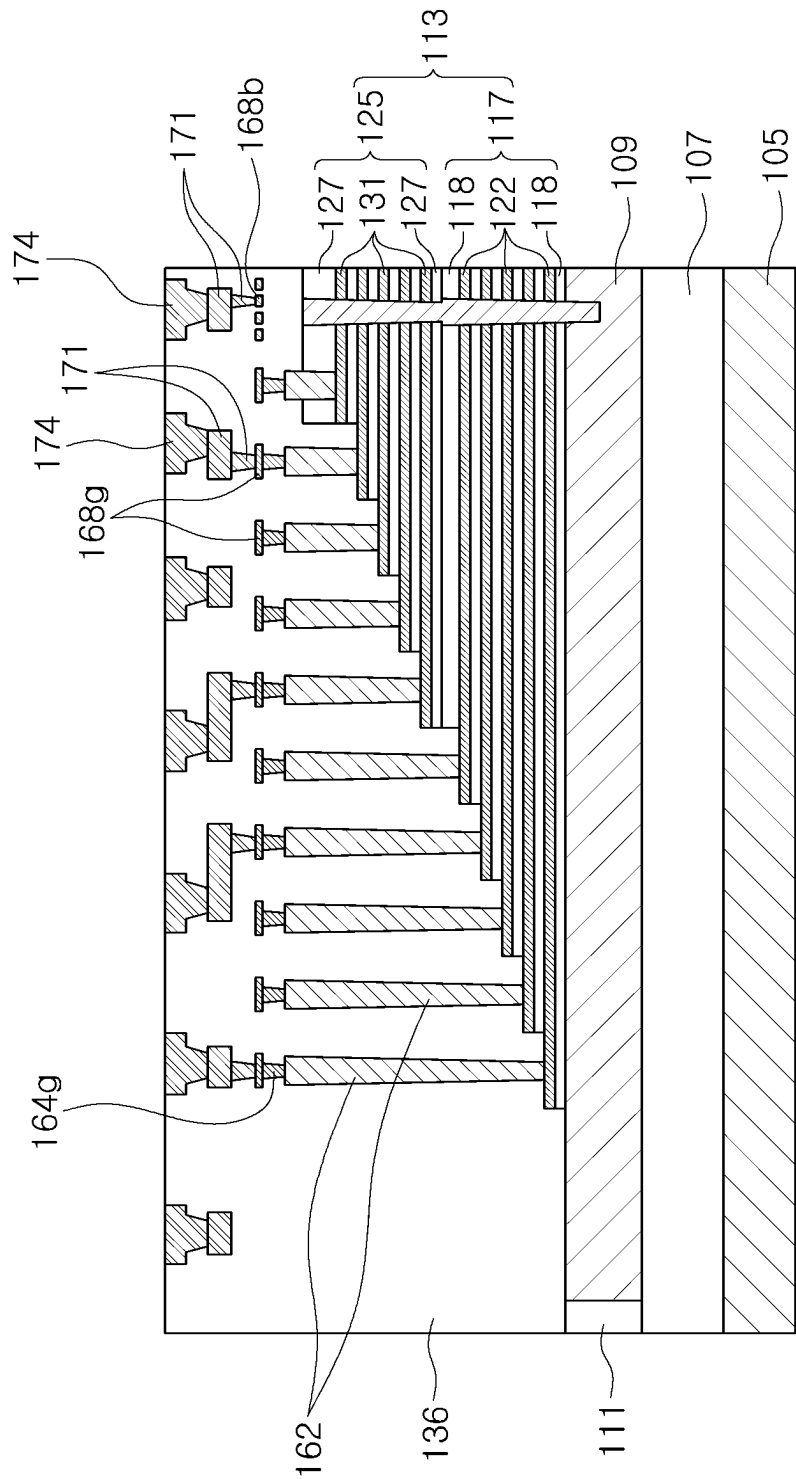
FIG. 13B



IVa-IVa'

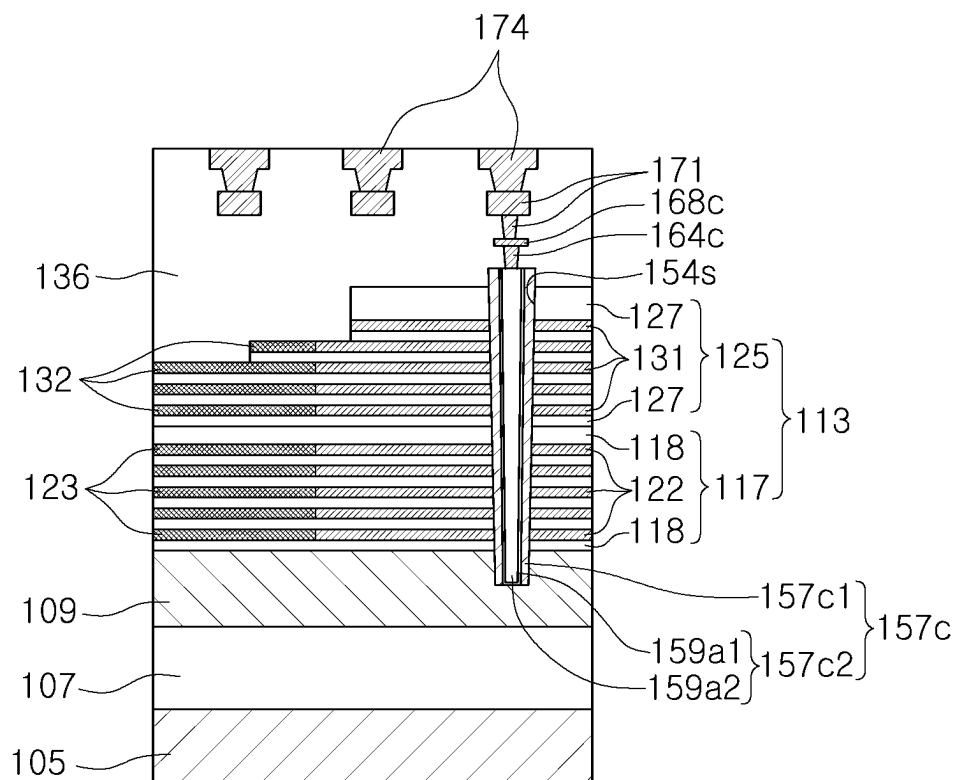
FIG. 13C





III-III'

FIG. 14B



IVa-IVa'

FIG. 14C

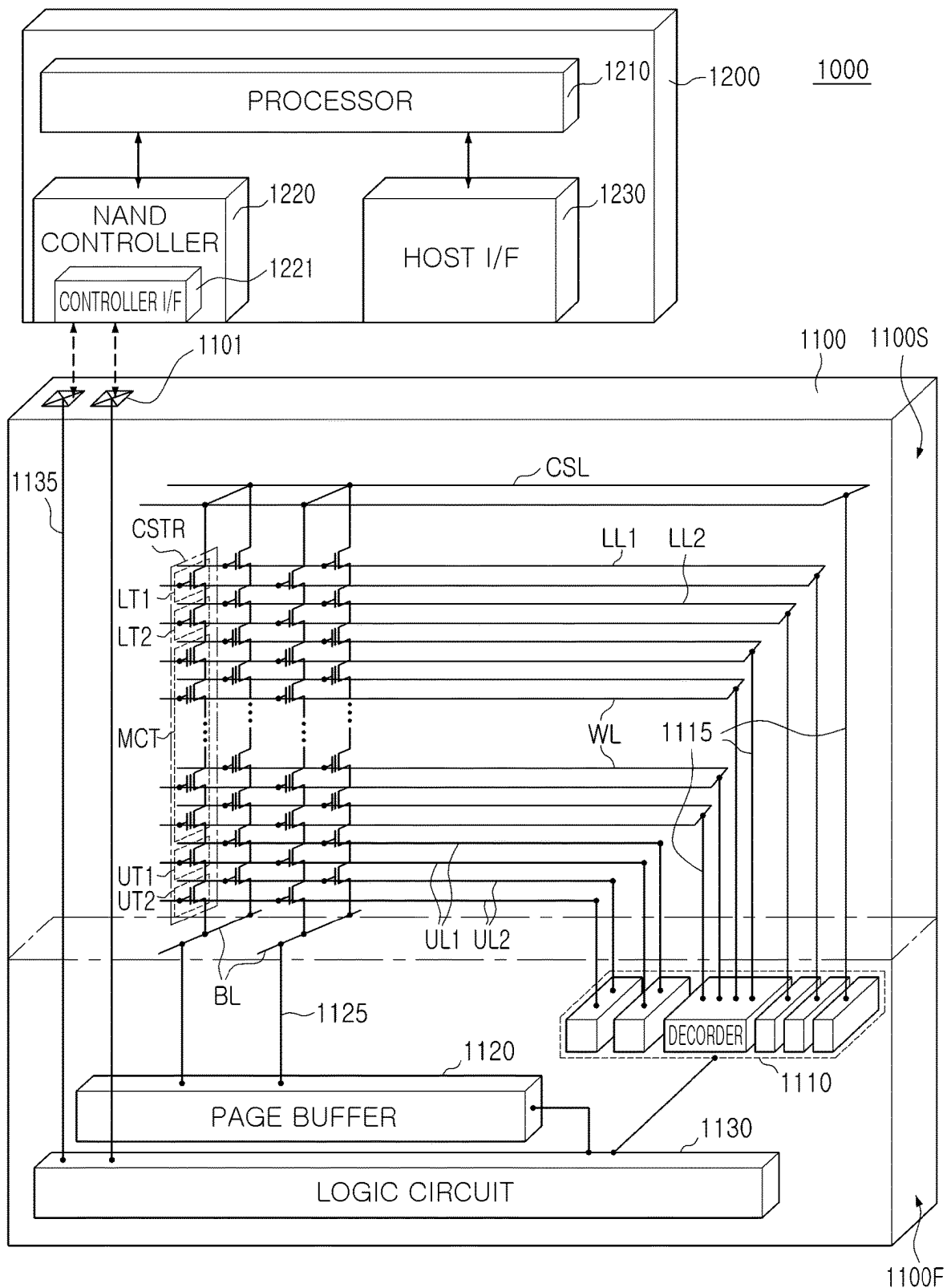


FIG. 15

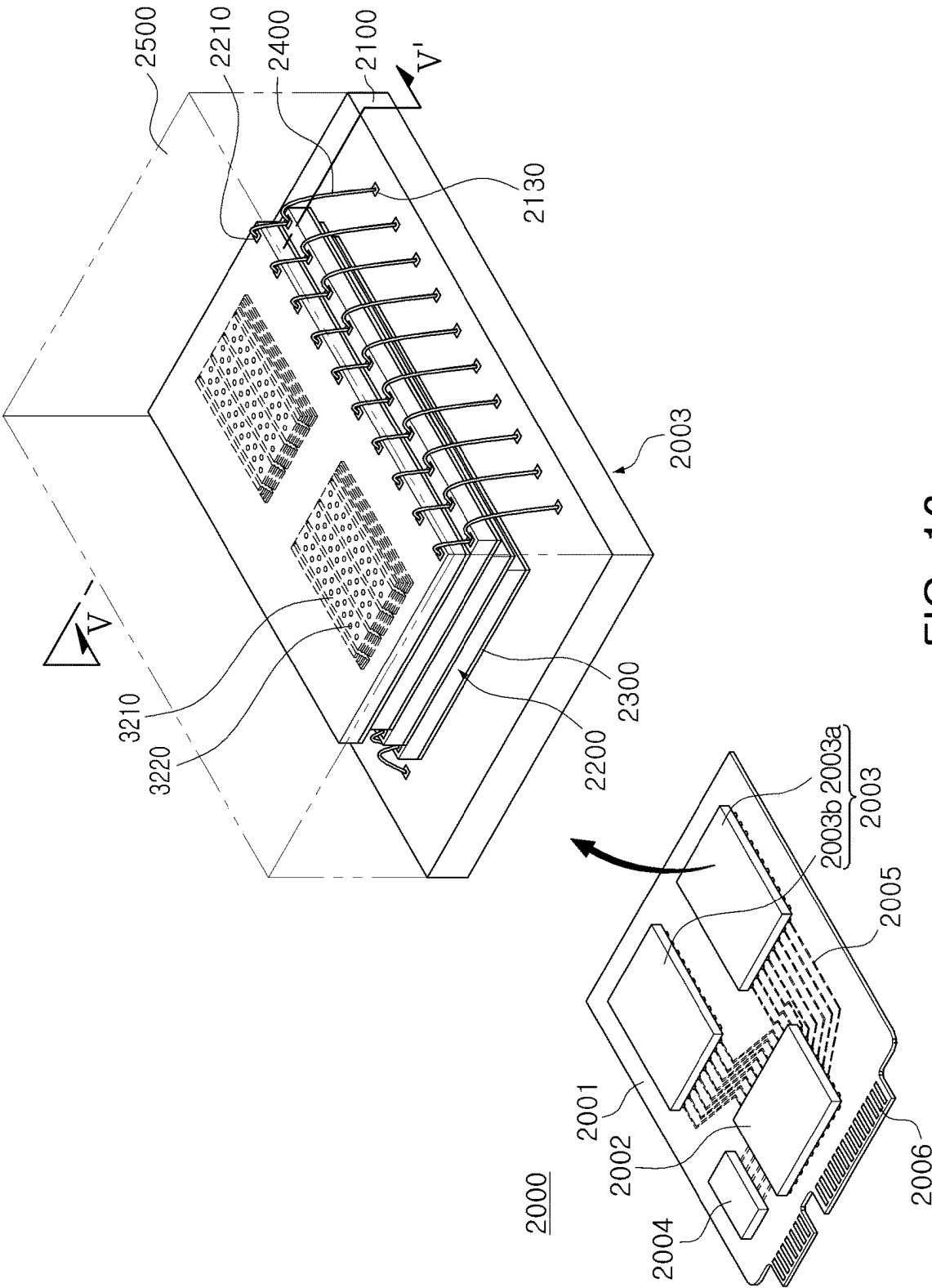
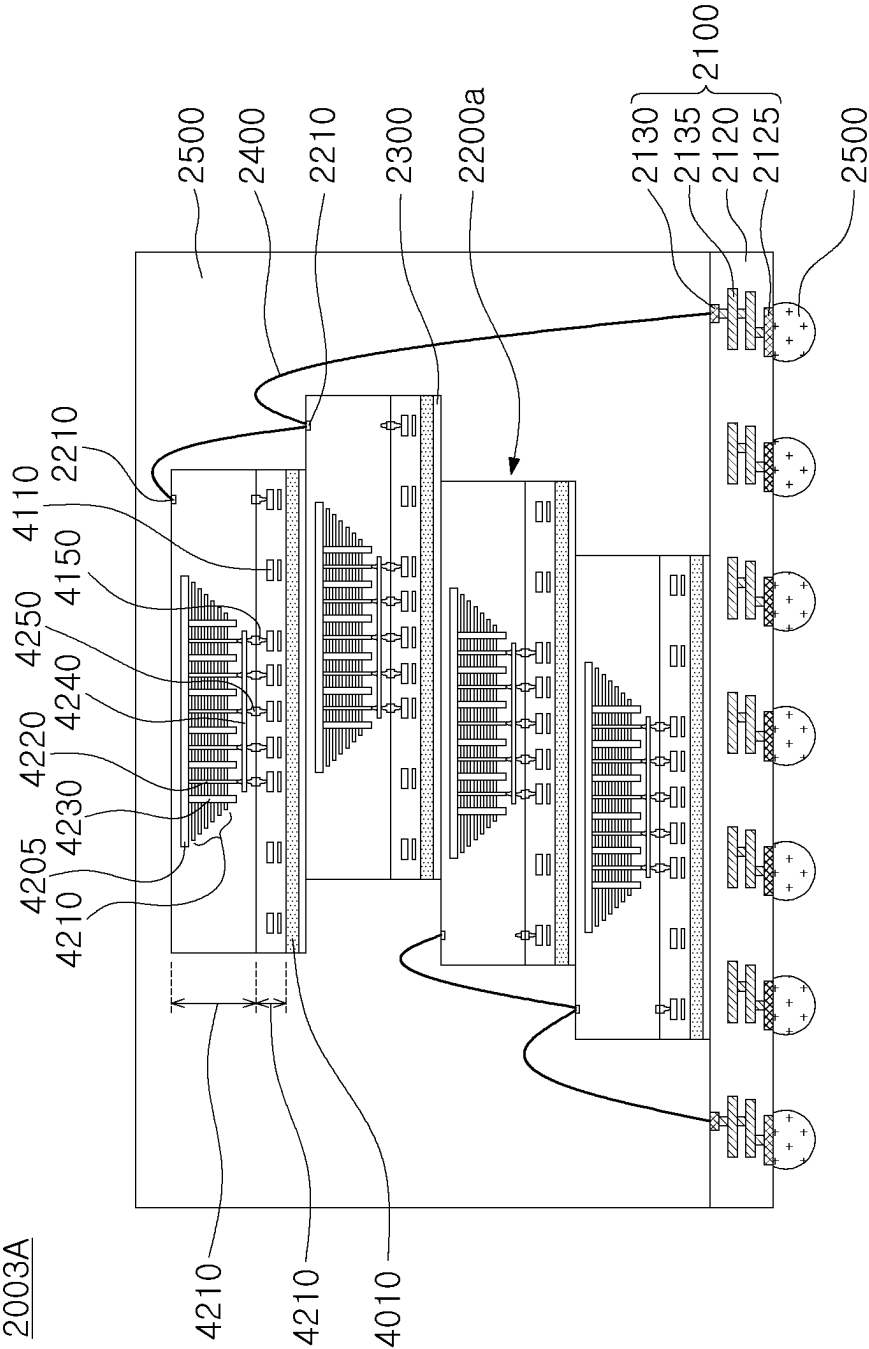


FIG. 16



V-V'

FIG. 17

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SEMICONDUCTOR DEVICE AND DATA STORAGE SYSTEM INCLUDING THE SAME

CROSS-REFERENCE TO RELATED APPLICATION(S)

This application claims the benefit under 35 USC 119(a) of Korean Patent Application No. 10-2020-0151779 filed on Nov. 13, 2020, in the Korean Intellectual Property Office, the entire contents of which are incorporated herein by reference for all purposes.

BACKGROUND

1. Field

The present inventive concept relates to a semiconductor device and a data storage system including the same.

2. Description of Related Art

An electronic system required to store data may require a semiconductor device capable of storing high-capacity data. Accordingly, research is being conducted on a method of increasing the data storage capacity of the semiconductor device. For example, as one of the methods of increasing the data storage capacity of the semiconductor device, a semiconductor device including memory cells arranged in three dimensions instead of memory cells arranged in two dimensions has been proposed.

SUMMARY

Example embodiments provide a semiconductor device having improved integration and reliability.

Example embodiments provide a data storage system including the semiconductor device.

According to example embodiments, a semiconductor device includes: a first structure including a peripheral circuit; and a second structure disposed on the first structure and bonded to the first structure, wherein the second structure includes: a pattern structure; an upper insulating layer disposed on the pattern structure; a stack structure disposed between the first structure and the pattern structure and including a first stack portion and a second stack portion spaced apart from each other in a horizontal direction, the first and second stack portions respectively including horizontal conductive layers and interlayer insulating layers alternately stacked in a vertical direction; separation structures penetrating through the stack structure and separating the stack structure; memory vertical structures penetrating through the first stack portion of the stack structure; and a contact structure penetrating through the second stack portion, the pattern structure, and the upper insulating layer. The contact structure includes a lower contact plug penetrating through at least the second stack portion of the stack structure and an upper contact plug in contact with the lower contact plug and extending upwardly to penetrate through the pattern structure and the upper insulating layer.

According to example embodiments, a semiconductor device includes: a first structure including a peripheral circuit; and a second structure disposed on the first structure and bonded to the first structure, wherein the second structure includes: a pattern structure; an upper insulating layer disposed on the pattern structure; word lines disposed between the pattern structure and the first structure, and stacked in a vertical direction; dummy horizontal conduc-

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tive layers disposed between the pattern structure and the first structure, and stacked in the vertical direction; memory vertical structures penetrating through the word lines in the vertical direction and in contact with the pattern structure, separation structures penetrating through the word lines in the vertical direction and in contact with the pattern structure; and a contact structure penetrating through the dummy horizontal conductive layers, the pattern structure, and the upper insulating layer in the vertical direction. The dummy horizontal conductive layers are electrically isolated.

According to example embodiments, a data storage system includes a semiconductor device including a first structure including a peripheral circuit and a second structure disposed on the first structure and including a data storage layer storing data; and a controller electrically connected to the semiconductor device, wherein the second structure includes: a pattern structure; an upper insulating layer disposed on the pattern structure; word lines disposed between the pattern structure and the first structure, and stacked in a vertical direction; dummy horizontal conductive layers disposed between the pattern structure and the first structure, and stacked in the vertical direction; memory vertical structures penetrating through the word lines in the vertical direction and in contact with the pattern structure; separation structures penetrating through the word lines in the vertical direction and in contact with the pattern structure; and a contact structure penetrating through the dummy horizontal conductive layers, the pattern structure, and the upper insulating layer in the vertical direction, and wherein the dummy horizontal conductive layers are electrically isolated.

BRIEF DESCRIPTION OF DRAWINGS

The above and other aspects, features, and advantages of the present inventive concept will be more clearly understood from the following detailed description, taken in conjunction with the accompanying drawings, in which like numerals refer to like elements. In the drawings:

FIGS. 1, 2, and 3A to 3D are schematic views each illustrating an example of a semiconductor device, according to an example embodiment;

FIG. 4A is a partially enlarged cross-sectional view illustrating a modified example of the semiconductor device, according to an example embodiment;

FIG. 4B is a partially enlarged cross-sectional view illustrating a modified example of the semiconductor device, according to an example embodiment;

FIG. 4C is a partially enlarged cross-sectional view illustrating a modified example of the semiconductor device, according to an example embodiment;

FIG. 4D is a partially enlarged cross-sectional view illustrating a modified example of the semiconductor device, according to an example embodiment;

FIG. 5A is a partially enlarged cross-sectional view illustrating an example of the semiconductor device, according to an example embodiment;

FIG. 5B is a partially enlarged cross-sectional view illustrating a modified example of the semiconductor device, according to an example embodiment;

FIG. 6 is a plan view illustrating a modified example of a component of the semiconductor device, according to an example embodiment;

FIGS. 7 and 8 are views illustrating a modified example of the semiconductor device, according to an example embodiment;

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FIG. 9 is a perspective view illustrating another modified example of the semiconductor device, according to an example embodiment;

FIG. 10 is a cross-sectional view illustrating a modified example of the semiconductor device, according to an example embodiment;

FIGS. 11, 12A to 12C, 13A to 13C, and 14A to 14C are cross-sectional views each illustrating an example of a method of manufacturing the semiconductor device, according to an example embodiment;

FIG. 15 is a schematic view illustrating a data storage system including a semiconductor device, according to an example embodiment;

FIG. 16 is a schematic perspective view illustrating the data storage system including the semiconductor device, according to an example embodiment; and

FIG. 17 is a schematic cross-sectional view illustrating the data storage system including the semiconductor device, according to an example embodiment.

DETAILED DESCRIPTION

Hereinafter, example embodiments will be described with reference to the accompanying drawings.

Hereinafter, terms such as 'upper', 'upper portion', 'upper surface', 'lower', 'lower portion', 'lower surface' and 'side surface' are indicated by reference numerals and may be understood as referring to drawings, unless otherwise indicated.

FIG. 1 is a schematic perspective view illustrating a semiconductor device according to an example embodiment; FIG. 2 is an enlarged plan view illustrating a region indicated by "A" in FIG. 1; FIG. 3A is a cross-sectional view illustrating a region taken along lines I-I' and II-II' of FIG. 2; FIG. 3B is a cross-sectional view illustrating a region taken along line III-III' of FIG. 1; FIG. 3C is a cross-sectional view illustrating a region taken along line IV-IV' of FIG. 1; and FIG. 3D is an enlarged cross-sectional view illustrating a portion indicated by "B" in FIG. 3A.

Referring to FIGS. 1, 2, and 3A to 3D, a semiconductor device 1 according to an example embodiment may include a first structure 3 and a second structure 103 disposed on the first structure 3. In some embodiments, the first structure 3 may be disposed on and bonded to the second structure 103.

The first structure 3 may be a first semiconductor chip including a peripheral circuit, and the second structure 103 may be a second semiconductor chip including a memory cell array region in which memory cells capable of storing data are arranged in three dimensions.

In an example embodiment, the first structure 3 may include: a semiconductor substrate 6; an isolation region 9s disposed on the semiconductor substrate 6 and defining a peripheral active region 9a; a peripheral circuit 12 formed on the semiconductor substrate 6; first bonding pads 18 electrically connected to the peripheral circuit 12; and a first insulating structure 21 disposed on the semiconductor substrate 6, covering the peripheral circuit 12 and having an upper surface coplanar with upper surfaces of the first bonding pads 18. Terms such as "same," "equal," "planar," or "coplanar," as used herein, encompass near identity including variations that may occur, for example, due to manufacturing processes. The term "substantially" may be used herein to emphasize this meaning, unless the context or other statements indicate otherwise.

The peripheral circuit 12 may include a circuit device 14 such as a transistor including a peripheral gate 14a and a peripheral source/drain 14b, and a circuit interconnection 16

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electrically connected to the circuit device 14. The circuit device 14 may further include a circuit element such as a resistor and a capacitor in addition to an active element such as the transistor.

The circuit interconnection 16 may be electrically connected to the first bonding pads 18. Therefore, the circuit interconnection 16 may electrically connect the first bonding pads 18 and the peripheral circuit 12 to each other.

In an example embodiment, the first bonding pads 18 may include a metal, for example, copper.

In an example embodiment, the second structure 103 may include a pattern structure 109, a stack structure 113, separation structures 157s, memory vertical structures 143c, a contact structure 190, second insulating structures 136, 111, and 177, and second bonding pads 174.

The pattern structure 109 may include a doped silicon layer. For example, the pattern structure 109 may include a polysilicon layer having an N-type conductivity.

The stack structure 113 may be disposed between the pattern structure 109 and the first structure 3.

In an example embodiment, the stack structure 113 may include a plurality of stacked groups 113Ga, 113Gb, and 113D. For example, the plurality of stacked groups 113Ga, 113Gb, and 113D of the stack structure 113 may include the first stacked group 113Ga and the second stacked group 113Gb and the dummy stacked group 113D, which are spaced apart from one another in a horizontal direction. For example, the first stacked group 113Ga and the second stacked group 113Gb may be spaced apart from each other in a first horizontal direction X. The dummy stacked group 113D may be spaced apart from the first stacked group 113Ga and the second stacked group 113Gb in a second horizontal direction Y perpendicular to the first horizontal direction X. However, the numbers and arrangement positions of the first stacked group 113Ga, the second stacked group 113Gb and the dummy stacked group 113D are not limited to the shape illustrated in FIG. 1 and may be variously modified.

In an example embodiment, each of the first and second stacked groups (e.g., first and second stacked groups 113Ga and 113Gb in FIG. 1) may include a first flat region 114/1 and a first stepped region 114s1 disposed on at least one side of the first flat region 114/1. The first stepped region 114s1 may surround the first flat region 114/1.

In an example embodiment, the dummy stacked group (e.g., the dummy stacked group 113D in FIG. 1) may include a second flat region 114/2 and a second stepped region 114s2 disposed on at least one side of the second flat region 114/2. The second stepped region 114s2 may surround the second flat region 114/2.

In an example embodiment, each of the first stacked group 113Ga and the second stacked group 113Gb may include a plurality of first stack portions 113a.

In an example embodiment, at least one of the first stacked group 113Ga and the second stacked group 113Gb may include one or more second stack portions 113b. Accordingly, the stack structure 113 may include the first stack portions 113a and the one or more the second stack portions 113b.

In an example embodiment, at least one of the second stack portions 113b may be disposed between the first stack portions 113a. For example, at least one of the first stacked group 113Ga and the second stacked group 113Gb may include the plurality of first stack portions 113a, and one second stack portion 113b may be disposed between two first stack portions 113a of the plurality of first stack portions 113a.

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In an example embodiment, the separation structures **157s** may be disposed in separation trenches **154s** penetrating through the stack structure **113** and separating the stack structure **113** in the first horizontal direction X. Accordingly, the separation structures **157s** may penetrate through the stack structure **113**. For example, the separation structures **157s** may penetrate through the first stacked group **113Ga** of the stack structure **113** and may penetrate through the second stacked group **113Gb** of the stack structure **113**. Each of the separation structures **157s** may have a shape of a line extending in the first horizontal direction X. The separation structures **157s** may extend into the pattern structure **109** and may be in contact with the pattern structure **109**. The term “contact,” as used herein, refers to a direct connection (i.e., touching) unless the context indicates otherwise.

Some of the separation structures **157s** may penetrate through the first stacked group **113Ga** and divide the first stacked group **113Ga** into the plurality of stack portions **113a** and **113b** spaced apart from each other in the second horizontal direction Y, and some of the separation structures **157s** may penetrate through the second stacked group **113Gb** and divide the second stacked group **113Gb** into the plurality of stack portions **113a** and **113b** spaced apart from each other in the second horizontal direction Y.

At least one of the plurality of stack portions **113a** and **113b** spaced apart from each other in the second horizontal direction Y may be the second stack portion **113b**, and the plurality of stack portions may be the first stack portions **113a**.

In an example embodiment, each of the first and second stack portions **113a** and **113b** of the stack structure **113** may include interlayer insulating layers **118** and **127** and horizontal gate layers **122** and **131**, which are alternately and repeatedly stacked in a vertical direction Z. Each of the horizontal gate layers **122** and **131** may include a conductive layer. The horizontal gate layers **122** and **131** may be stacked and spaced apart from each other in the vertical direction Z.

In an example embodiment, at least some of the horizontal gate layers **122** and **131** of the first stack portion **113a** may be word lines. Among the horizontal gate layers **122** and **131** of the first stack portion **113a**, one or more upper horizontal gate layers disposed at its upper portion and/or one or more lower horizontal gate layers disposed at its lower portion may be selection gate electrodes, and a plurality of horizontal gate layers disposed between the one or plurality of upper horizontal gate layers and the one or more lower horizontal gate layers may be the word lines.

In an example embodiment, the horizontal gate layers **122** and **131** of the second stack portion **113b** may be dummy horizontal conductive layers which may be electrically isolated.

In an example embodiment, the dummy stacked group **113D** of the stack structure **113** may include the interlayer insulating layers **118** and **127** and horizontal insulating layers **123** and **132** which are alternately and repeatedly stacked in the vertical direction.

In an example embodiment, each of the first and second stack portions **113a** and **113b** of the stack structure **113** may include a first stacked region **125** and a second stacked region **117** disposed on the first stacked region **125**. Here, the first stacked region **125** may be disposed between the second stacked region **117** and the first structure **3**.

The first stacked region **125** may include the first interlayer insulating layers **127** and the first horizontal gate layers **131** which are alternately and repeatedly stacked in the vertical direction Z. A lowermost layer and an uppermost layer among the first interlayer insulating layers **127** and the

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first horizontal gate layers **131** may each be one of the first interlayer insulating layers **127**.

In an example embodiment, the second stacked region **117** may include the second interlayer insulating layers **118** and the second horizontal gate layers **122** which are alternately and repeatedly stacked in the vertical direction. A lowermost layer and an uppermost layer among the second interlayer insulating layers **118** and the second horizontal gate layers **122** may each be one of the second interlayer insulating layers **118**.

The first and second horizontal gate layers **131** and **122** may have pad regions GP arranged in a step shape within the first stepped region **114s1**. The pad regions GP of the first and second horizontal gate layers **131** and **122** may have a shape of the step down in a direction from the first structure **3** toward the pattern structure **109**. The pad regions GP may face the first structure **3**.

The second structure **103** may further include gate contact plugs **162** in contact with the pad regions GP and electrically connected to the first and second horizontal gate layer **131** and **122**. For example, the gate contact plugs **162** may be electrically connected to the first and second horizontal gate layers **131** and **122**, which may be the selection gate electrodes and the word lines, among the first and second horizontal gate layers **131** and **122**. The gate contact plugs **162** may extend downward from a portion of the first and second horizontal gate layer **131** or **122** in contact with the pad regions GP. For example, the gate contact plugs **162** may extend from the pad regions GP toward the first structure **3**. The gate contact plugs **162** may be formed of a conductive material.

The memory vertical structures **143c** may penetrate through the first stack portion **113a** of the stack structure **113**. For example, the memory vertical structures **143c** may penetrate through the horizontal gate layers **122** and **131**, which may be the selection gate electrodes and word lines of the first stack portion **113a**, in the vertical direction Z.

In an example embodiment, the memory vertical structures **143c** may extend into the pattern structure **109** from its portion penetrating through the first stack portion **113a** of the stack structure **113** to be in contact with the pattern structure **109**.

The second structure **103** may further include dummy vertical structures **143d**. The dummy vertical structures **143d** may penetrate through the second stack portion **113b** of the stack structure **113** and may be in contact with the pattern structure **109**.

In an example embodiment, the dummy vertical structures **143d** may include the same material layer as the memory vertical structures **143c**. For example, the dummy vertical structures **143d** may be formed by substantially the same process as the memory vertical structures **143c**, and may have a cross-sectional structure substantially the same as the memory vertical structures **143c**.

In another example embodiment, the dummy vertical structures **143d** may be formed by a process different from the memory vertical structures **143c**. For example, each of the dummy vertical structures **143d** may be formed of a silicon oxide column.

The second bonding pads **174** may be in contact with and bonded to the first bonding pads **18**. The first and second bonding pads **18** and **174** may include the same conductive material, for example, copper.

In an example embodiment, the second insulating structures **136**, **111**, and **177** may include an outer insulating layer **111** disposed on a side surface of the pattern structure **109**, and an upper insulating layer **177** disposed on the pattern

structure 109 and the outer insulating layer 111. The outer insulating layer 111 may contact the side surface of the pattern structure 109, and the upper insulating layer 177 may contact upper surfaces of the pattern structure 109 and the outer insulating layer 111. The second insulating structures 136, 111, and 177 may further include a capping insulating layer 136 disposed on the first structure 3 and covering the stack structure 113 while surrounding side surfaces of the second bonding pads 174. The pattern structure 109 and the outer insulating layer 111 may contact an upper surface of the capping insulating layer 136. The upper insulating layer 177 may be formed of a silicon oxide layer. The capping insulating layer 136 may include a single material layer or multiple material layers. For example, the capping insulating layer 136 may be formed of a silicon oxide layer or may be formed to include a silicon oxide layer and a material having etching selectivity with the silicon oxide layer, for example, a silicon nitride layer.

The capping insulating layer 136 may fill a space between the stack structure 113 and the first structure 3 while surrounding each of the first stacked group 113Ga, the second stacked group 113Gb and the dummy stacked group 113D. For example, a portion of the capping insulating layer 136 may fill the space between the first stacked group 113Ga, the second stacked group 113Gb and the dummy stacked group 113D, which are spaced apart from one another.

A plurality of contact structures 190 may be arranged. At least some of the plurality of contact structures 190 may penetrate through the stack structure 113, the pattern structure 109, and the upper insulating layer 177. Each of the plurality of contact structures 190 may include a lower contact structure 157c and an upper contact structure 183c disposed on the lower contact structure 157c.

In the following, for easier understanding, the description is made focusing on one contact structure 190 including a portion penetrating through the stack structure 113.

The lower contact structure 157c of the contact structure 190 may be disposed in a lower contact hole 154c passing through the second stack portion 113b of the stack structure 113, and the upper contact structure 183c thereof may be disposed in an upper contact hole 180 passing through the pattern structure 109 and the upper insulating layer 177.

In an example embodiment, the upper contact structure 183c adjacent to the lower contact structure 157c may have a greater width than the lower contact structure 157c adjacent to the upper contact structure 183c.

The lower contact structure 157c may include a conductive lower contact plug 157c2 and an insulating lower spacer 157c1 surrounding a side surface of the lower contact plug 157c2.

In an example embodiment, the lower contact plug 157c2 of the lower contact structure 157c may include a portion penetrating through the second stack portion 113b of the stack structure 113 and a portion extending into the pattern structure 109 from the portion penetrating through the second stack portion 113b of the stack structure 113. Accordingly, an upper surface (e.g., the upper surface 157cU in FIG. 3D) of the lower contact plug 157c2 may be disposed at a height level between a lower surface (e.g., the lower surface 109L in FIG. 3) of the pattern structure 109 and an upper surface (e.g., the upper surface 109U in FIG. 3D) of the pattern structure 109. For example, the upper surface 157cU of the lower contact plug 157c2 may be disposed at a level higher than the lower surface 109L of the pattern structure 109, and may be disposed at a level lower than the upper surface 109U of the pattern structure 109. In an

example embodiment, a height difference between the upper surface 157cU of the lower contact plug 157c2 and the lower surface 109L of the pattern structure 109 may be smaller than a height difference between the upper surface 157cU of the lower contact plug 157c2 and the upper surface 109U of the pattern structure 109.

The lower contact plug 157c2 may include a first liner layer 159a1 and a first pillar pattern 159a2.

In an example embodiment, the first liner layer 159a1 may cover at least a side surface of the first pillar pattern 159a2. For example, the first liner layer 159a1 may contact at least the side surface of the first pillar pattern 159a2.

In an example embodiment, the first liner layer 159a1 may include a portion covering the side surface of the first pillar pattern 159a2 and a portion covering an upper surface of the first pillar pattern 159a2. For example, the first liner layer 159a1 may include a portion contacting the side surface of the first pillar pattern 159a2 and a portion contacting an upper surface of the first pillar pattern 159a2. Here, the "upper surface" of the first pillar pattern 159a2 may be a term referred to with reference to FIG. 3A.

The upper contact structure 183c may include an upper contact plug 183c2 in contact with the lower contact plug 157c2 and an insulating upper spacer 183c1 surrounding at least a side surface of the upper contact plug 183c2. The insulating upper spacer 183c1 may contact at least the side surface of the upper contact plug 183c2.

The upper contact plug 183c2 may be in contact with the upper surface 157cU of the lower contact plug 157c2 and may be in contact with the side surface of the lower contact plug 157c2, which is adjacent to the upper surface 157cU of the lower contact plug 157c2.

The upper spacer 183c1 may be formed of an insulating material such as silicon oxide and/or silicon nitride. The upper contact plug 183c2 may include at least one of a conductive material, for example, a metal nitride (e.g., titanium nitride (TiN), tantalum nitride (TaN), or tungsten nitride (WN)) or a metal (e.g., tungsten (W), copper (Cu), or aluminum (Al)).

In an example embodiment, a thickness of the upper spacer 183c1 in the horizontal direction may be greater than a thickness of the lower spacer 157c1 in the horizontal direction.

In an example embodiment, a lower surface 183cL of the upper contact plug 183c2 may be disposed at a level higher than the lower surface 109L of the pattern structure 109, and at a level lower than the upper surface 109U of the pattern structure 109.

Each of the separation structures 157s may include an insulating separation spacer 157s1 and a conductive separation pattern 157s2. The separation pattern 157s2 may include a second liner layer 159b1 and a second pillar pattern 159b2. The second liner layer 159b1 may cover an upper surface of the second pillar pattern 159b2 while covering a side surface of the second pillar pattern 159b2. For example, the second liner layer 159b1 may contact the upper and side surfaces of the second pillar pattern 159b2. Here, the upper surface of the second pillar pattern 159b2 may be disposed at a height level between the lower surface 109L of the pattern structure 109 and the upper surface 109U of the pattern structure 109.

In an example embodiment, the separation spacer 157s1 of the separation structures 157s and the lower spacer 157c1 of the lower contact structure 157c may be formed of the same material, for example, silicon oxide and/or silicon nitride, formed by the same process.

In an example embodiment, the separation pattern **157s2** of the separation structures **157s** and the lower contact plug **157c2** of the lower contact structure **157c** may be formed of the same conductive material and formed by the same process. For example, the first liner layer **159a1** of the lower contact plug **157c2** and the second liner layer **159b1** of the separation pattern **157s2** may include the same conductive material, for example, metal nitride (e.g., TiN, TaN, or WN), and be formed by the same process, and the first pillar pattern **159a2** of the lower contact plug **157c2** and the second pillar pattern **159b2** of the separation pattern **157s2** may include the same conductive material, for example, metal (e.g., W), and be formed by the same process. For another example, the lower contact plug **157c2** and the separation pattern **157s2** may include at least one of doped silicon, a metal-semiconductor compound, a metal nitride, and a metal.

The second structure **103** disposed between the stack structure **113** and the first structure **3** may further include bit lines **168b**, gate interconnections **168g**, a contact interconnection **168c**, bit line connection vias **164b**, gate connection vias **164g**, and a contact connection via **164c**. The gate interconnections **168g** and the contact interconnection **168c** may each be referred to as the gate interconnection lines **168g** and the contact interconnection lines **168c**. The bit line connection vias **164b** may be disposed between the bit lines **168b** and the memory vertical structures **143c**, and may electrically connect the bit lines **168b** and the memory vertical structures **143c** to each other. The bit lines **168b** may not be electrically connected to the dummy vertical structures **143d**. The gate connection vias **164g** may be disposed between the gate interconnections **168g** and the gate contact plugs **162**, and may electrically connect the gate interconnections **168g** and the gate contact plugs **162** to each other. The contact connection via **164c** may be disposed between the contact interconnection **168c** and the lower contact plug **157c2**, and may electrically connect the contact interconnection **168c** and the lower contact plug **157c2** to each other. The second structure **103** may further include an interconnection structure **171** electrically connecting the bit lines **168b**, the gate interconnections **168g** and the contact interconnection **168c** with the second bonding pads **174**. The second structure **103** may further include a conductive pattern **193** disposed on the upper insulating layer **177** and electrically connected to the upper contact plug **183c2**.

In an example embodiment, the contact structure **190** may be an input/output contact structure for transmitting an input/output signal of the semiconductor device **1**.

Next, various modified examples of the contact structure **190** are respectively described with reference to FIGS. **4A** to **4D**. Each of FIGS. **4A** to **4D** is a partially enlarged cross-sectional view corresponding to the partially enlarged cross-sectional view of FIG. **3D**, and may indicate a modified example of the portion of the contact structure **190**, which is illustrated in FIG. **3D**. Hereinafter, the description is made focusing on a portion of the contact structure **190**, which may be modified, with reference to FIGS. **4A** to **4D**, respectively.

Referring to FIG. **4A**, the modified example may provide a lower contact structure **157ca** including a lower contact plug **157c2a** and the lower spacer **157c1**, and the lower contact plug **157c2a** may include a first pillar pattern **159a2a** and a first liner layer **159a1a** surrounding a portion of a side surface of the first pillar pattern **159a2a**. The first pillar pattern **159a2a** may extend into the upper contact plug **183c2** and may be in contact with the upper contact plug **183c2**. The upper contact plug **183c2** may be in contact with

the upper surface **157cU** of the first pillar pattern **159a2a**, and may be in contact with a side surface of a portion of the first pillar pattern **159a2a**, which extends into the upper contact plug **183c2**.

Referring to FIG. **4B**, another modified example may provide the upper contact structure **183ca** including an upper contact plug **183c2a** and an upper spacer **183c1a**, and the upper spacer **183c1a** may surround a side surface of the upper contact plug **183c2a**.

In an example embodiment, a lower surface **183cLa** of the upper contact plug **183c2a** may be disposed at substantially the same level as the lower surface **109L** of the pattern structure **109** or at a level lower than the lower surface **109L** of the pattern structure **109**.

Referring to FIG. **4C**, another modified example may provide the lower contact structure **157ca** substantially the same as that illustrated in FIG. **4A** and the upper contact structure **183ca** substantially the same as that described with reference to FIG. **4B**. For example, the upper surface and a portion of the side surface of the first pillar pattern **159a2a** of the lower contact structure **157ca** described with reference to FIG. **4A** may be in contact with the upper contact plug **183c2a**. A portion of the first liner layer **159a1a** of the lower contact structure **157ca** described with reference to FIG. **4A** may be interposed between the side surface of the first pillar pattern **159a2a** and the upper contact plug **183c2a**.

Referring to FIG. **4D**, another modified example may provide the upper contact structure **183ca** substantially the same as that described with reference to FIG. **4B**.

A lower contact structure **157cb** may include a lower contact plug **157c2b** and the lower spacer **157c1**, and the lower contact plug **157c2b** may include a first pillar pattern **159a2b** and a first liner layer **159a1b** surrounding at least a portion of a side surface of the first pillar pattern **159a2b**.

In an example embodiment, the first pillar pattern **159a2b** may extend into the upper contact plug **183c2** and may be in contact with the upper contact plug **183c2**.

In an example embodiment, an upper surface **157cU'** of the lower contact plug **157c2b** may be disposed at the same height level as the lower surface **109L** of the pattern structure **109**, or at a height level lower than the lower surface **109L** of the pattern structure **109**.

Next, the description describes an example embodiment of the memory vertical structures **143c**. FIG. **5A** is an enlarged cross-sectional view illustrating a region indicated by "C" in FIG. **3A**. Hereinafter, referring to FIG. **5A**, the description is made focusing on any one of the memory vertical structures **143c** and describes example embodiments of the horizontal gate layers **122** and **131** and the pattern structure **109**.

Referring to FIGS. **3A** and **5A**, in an example embodiment, the pattern structure **109** may include a plurality of layers. For example, the pattern structure **109** may include a first pattern layer **109a**, a second pattern layer **109b** disposed under and contacting the first pattern layer **109a**, and a third pattern layer **109c** disposed under and contacting the second pattern layer **109b**. At least one of the first pattern layer **109a**, the second pattern layer **109b**, and the third pattern layer **109c** may include a polysilicon layer. For example, each of the first pattern layer **109a**, the second pattern layer **109b**, and the third pattern layer **109c** may include the polysilicon layer. At least one of the first pattern layer **109a**, the second pattern layer **109b**, and the third pattern layer **109c** may include the polysilicon layer having the N-type conductivity.

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The memory vertical structures **143c** may penetrate through the stack structure **113** and extend into the pattern structure **109**.

The memory vertical structures **143c** may include a core region **149**, a channel layer **147**, a pad pattern **151**, and a data storage structure **145**.

The channel layer **147** may cover the side surface and bottom surface of the core region **149**. The channel layer **147** may be formed of a semiconductor material such as silicon. For example, the channel layer **147** may be formed of polysilicon.

The core region **149** may be formed of silicon oxide or silicon oxide having a void or a seam therein.

The pad pattern **151** may be disposed on the core region **149** and in contact with the core region **149** and the channel layer **147**. The pad pattern **151** may be formed of polysilicon having the N-type conductivity.

In an example embodiment, the data storage structure **145** may cover an upper surface of the channel layer **147** by covering an outer side surface of the channel layer **147**. Here, the upper surface of the channel layer **147** may be the upper surface of the channel layer **147** as viewed with reference to FIG. 5A.

In an example embodiment, the second pattern layer **109b** may penetrate through the data storage structure **145** and be in contact with the channel layer **147**, and the first and third pattern layers **109a** and **109c** may be spaced apart from the channel layer **147** by the data storage structure **145**. The data storage structure **145** may include a first data storage structure **145U** disposed at its lower portion and a second data storage structure **145L** disposed at its upper portion. The first data storage structure **145U** may be separated from the second data storage structure **145L** by the second pattern layer **109b**.

The data storage structure **145** may include a first dielectric layer **145a**, a second dielectric layer **145c**, and a data storage layer **145b** disposed between the first dielectric layer **145a** and the second dielectric layer **145c**. At least one of the first and second dielectric layers **145a** and **145c** may include silicon oxide and/or a high-k dielectric.

In an example embodiment, the data storage layer **145b** may include a material capable of trapping an electric charge, for example, silicon nitride. The data storage layer **145b** may be a charge trap layer.

In an example embodiment, the data storage layer **145b** may include regions capable of storing data in a semiconductor device such as a NAND flash memory device. For example, the data storage layer **145b** may include the data storage regions capable of storing data between the gate layer, which may be the word lines among the horizontal gate layers **122** and **131**, and the channel layer **147**. Such data storage regions may each configure memory cells capable of storing data and may be arranged in one memory vertical structures **143c** in a substantially vertical direction, and the plurality of memory vertical structures **143c** including such data storage regions may be arranged in the horizontal direction. Accordingly, the plurality of memory vertical structures **143c** including the plurality of data storage regions capable of configuring the memory cells may be disposed, and the semiconductor device **1** may thus include a memory cell array region including the memory cells arranged in the three dimensions. Here, the memory cell array region may be defined as regions of the first and second stacked groups **113Ga** and **113Gb** in which the memory vertical structures **143c** are disposed. For another example, the memory vertical structures **143c** may include a data storage region of a memory device that stores data

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using a change in resistance. For example, the memory vertical structures **143c** may include a data storage structure of Resistive RAM (ReRAM) including any one of silicon oxide (SiOx), aluminum oxide (AlOx), magnesium oxide (MgOx), zirconium oxide (ZrOx), hafnium oxide (HfOx), silicon nitride (SiNx), tungsten oxide (WOx), and titanium oxide (TiOx), or a composite material including at least two of those materials. Alternatively, the memory vertical structures **143c** may include a data storage structure of a PRAM including a phase change memory material such as a chalcogenide material including germanium (Ge), antimony (Sb), and/or tellurium (Te).

In the example embodiments, the memory vertical structures **143c** may be referred to as a vertical structure, a vertical pattern, or the like.

Each of the horizontal gate layers **122** and **131** may include a first layer **133a** and a second layer **133b**. The first layer **133a** may cover the upper and lower surfaces of the second layer **133b** and extend to a space between the memory vertical structures **143c** and a side surface of the second layer **133b**.

In an example embodiment, the first layer **133a** may include a dielectric material, and the second layer **133b** may include a conductive material. For example, the first layer **133a** may include a high-k dielectric such as AlO, and the second layer **133b** may include a conductive material such as titanium nitride (TiN), tungsten nitride (WN), titanium (Ti), or tungsten (W).

For another example, the first layer **133a** may include a first conductive material (e.g., titanium nitride (TiN) or tungsten (W)), and the second layer **133b** may include a second conductive material (e.g., titanium (Ti) or tungsten (W)) different from the first conductive material.

For another example, each of the horizontal gate layers **122** and **131** may be formed of doped polysilicon, a metal-semiconductor compound (e.g., titanium silicide (TiSi), tantalum silicide (TaSi), cobalt silicide (CoSi), nickel silicide (NiSi), or tungsten silicide (WSi)), a metal nitride (e.g., titanium nitride (TiN), tantalum nitride (TaN), or tungsten nitride (WN)) or a metal (e.g., titanium (Ti) or tungsten (W)).

As described above, the stack structure **113** may include the first stacked region **125** and the second stacked region **117**.

Each of the memory vertical structures **143c** may include a lower vertical portion **145s2** penetrating through the first stacked region **125** and an upper vertical portion **145s1** penetrating through the second stacked region **117**.

In an example embodiment, a side surface of the lower vertical portion **145s2** and a side surface of the upper vertical portion **145s1** adjacent to each other may not be aligned in the vertical direction. Accordingly, a side surface of the memory vertical structures **143c** may have a bent portion in a region between the first horizontal conductive layers **131** of the first stacked region **125** and the second horizontal gate layers **122** of the second stacked region **117**. For example, a portion of the side surface of the memory vertical structures **143c** may extend in a horizontal direction, and may be coplanar with a surface of one of the second interlayer insulating layers **118**.

In an example embodiment, the upper vertical portion **145s1** of the memory vertical structures **143c**, which is adjacent to the lower vertical portion **145s2**, may have a width greater than the lower vertical portion **145s2** thereof, which is adjacent to the upper vertical portion **145s1**.

Next, a modified example of the memory vertical structures **143c** is described with reference to FIG. 5B. FIG. 5B

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is a partially enlarged view corresponding to the partially enlarged view of FIG. 5A. Hereinafter, the description is made focusing on the modified portion of the memory vertical structures **143c** described with reference to FIG. 5A.

Referring to FIG. 5B, a memory vertical structures **143c'** penetrating through the stack structure **113** and extending into the pattern structure **109** may include: an epitaxial channel layer **144** including a portion disposed in the pattern structure **109**, disposed at a level lower than a lower surface of at least the lowermost horizontal gate layer **122L** and having an upper surface disposed at a level higher than an upper surface of the next-lowest horizontal gate layer, among the horizontal gate layers **122** and **131**; a core region **149'** disposed under the epitaxial channel layer **144**; a channel layer **147'** interposed between the core region **149'** and the epitaxial channel layer **144** and covering a side surface of the core region **149'**; and a data storage structure **145'** covering an outer side surface of the channel layer **147'**. The data storage structure **145'** may include the first dielectric layer **145a**, the second dielectric layer **145c**, and the data storage layer **145b** disposed between the first and second dielectric layers **145a** and **145c**.

In an example embodiment, a dielectric layer **152** may be disposed between the lowermost horizontal gate layer **122L** and the epitaxial channel layer **144**.

In the example embodiments, the contact structure **190** may have a circular shape when viewed as a plane as illustrated in FIG. 1. However, the planar shape of the contact structure **190** is not limited to the circular shape. The description describes such a modified example of the planar shape of the contact structure **190** with reference to FIG. 6. FIG. 6 is a plan view illustrating a modified example of the planar shape of the contact structure.

Referring to FIG. 6, a contact structure **190'** may have an elongated shape in one direction, for example, an oval shape, a rectangular shape, a bar shape, or a line shape. For example, the lower contact structure **157c** of the contact structure **190'**, which includes the lower contact plug **157c2** and the lower spacer **157c1** may have the elongated shape in one direction, and the upper contact structure **183c** thereof, which includes the upper contact plug **183c2** and the upper spacer **183c1** may have the elongated shape in one direction. In example embodiments, the lower contact structure **157c** and the upper contact structure **183c** may both have shapes that are elongated in the one direction.

Next, a modified example of the semiconductor device **1** according to an example embodiment is described with reference to FIGS. 7 and 8. FIG. 7 is a schematic perspective view illustrating the modified example of the semiconductor device according to an example embodiment; and FIG. 8 is a schematic cross-sectional view illustrating a region taken along line IVa-IVa' of FIG. 7. Hereinafter, the description is made focusing on a portion modified from the semiconductor device **1** described with reference to FIGS. 1 through 3D.

Referring to FIGS. 7 and 8, a semiconductor device **1a** according to an example embodiment may include the first structure **3** and a second structure **103a**. The first structure **3** may be substantially the same as that described with reference to FIGS. 1 through 3D. The second structure **103a** may include first and second gate groups **113Ga** and **113Gb** including the first stack portions **113a** without including the second stack portions (e.g., the second stack portions **113b** in FIG. 1), instead of the first and second stack groups **113Ga** and **113Gb** including the first stack portions (e.g., the first stack portions **113a** in FIG. 1) and the second stack portions (e.g., the second stack portions **113b** in FIG. 1) described with reference to FIG. 1.

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The second structure **103a** may include a dummy stacked group **113D'** including a second stack portion **113b'**, instead of the dummy stacked group **113D** described with reference to FIGS. 1 through 3D.

The dummy stacked group **113D'** may further include a dummy stack portion **113d** in contact with the second stack portion **113b'**.

The second stack portion **113b'** of the dummy stacked group **113D'** may be formed of layers substantially the same as the first and second stack portions **113a** and **113b** of the first and second stacked groups **113Ga** and **113Gb** described with reference to FIGS. 1 to 3D. For example, the second stack portion **113b'** of the dummy stacked group **113D'** may include the interlayer insulating layers **118** and **127** and horizontal gate layers **122'** and **131'**, which are alternately and repeatedly stacked on each other. The dummy stack portion **113d** of the dummy stacked group **113D'** may include the interlayer insulating layers **118** and **127** and horizontal insulating layers **123** and **132**, which are alternately and repeatedly stacked on each other.

The second stack portion **113b'** of the dummy stacked group **113D'** may include a first stacked region **125'** and a second stacked region **117'** disposed on the first stacked region **125'**. The first stacked region **125'** may include the first interlayer insulating layers **127** and the first horizontal gate layers **131'** which are alternately and repeatedly stacked on each other, and the second stacked region **117'** may include the second interlayer insulating layers **118** and the second horizontal gate layers **122'** which are alternately and repeatedly stacked on each other.

The second structure **103a** may include a contact structure **290** having the same structure as the contact structure **190** described with reference to FIGS. 1 to 3D. For example, the contact structure **290** may include the lower contact structure **157c** penetrating through the second stack portion **113b'** of the dummy stacked group **113D'** and the upper contact structure **183c** penetrating through the pattern structure **109** and the upper insulating layer **177**. Here, the lower contact structure **157c** and the upper contact structure **183c** may be substantially the same as the structures of the lower contact structure **157c** and the upper contact structure **183c** described with reference to FIGS. 1 to 3D.

In an example embodiment, the contact structure **290** may be modified like any one of the modified examples of the contact structure **190** described with reference to FIGS. 4A to 4D and 6.

Next, another modified example of the semiconductor device **1** according to an example embodiment is described with reference to FIG. 9. FIG. 9 is a schematic perspective view illustrating the modified example of the semiconductor device according to an example embodiment. Hereinafter, the description is made focusing on a portion modified from the semiconductor devices **1** and **1a** described with reference to FIGS. 1 through 8.

Referring to FIG. 9, a semiconductor device **1b** according to an example embodiment may include the first structure **3** and a second structure **103b**. The first structure **3** may be substantially the same as that described with reference to FIGS. 1 to 3D. The second structure **103b** may include: the first and second stack groups **113Ga** and **113Gb** including the first stack portions (e.g., the first stack portions **113a** in FIG. 1) and the second stack portions (e.g., the second stack portions **113b** in FIG. 1) described with reference to FIG. 1; the dummy stacked group **113D'** including the second stack portion **113b'** described with reference to FIG. 7; the contact

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structure **190** described with reference to FIGS. **1** to **3D**; and the contact structure **290** described with reference to FIGS. **7** and **8**.

In an example embodiment, the contact structures **190** and **290** may each be modified like any one of the modified examples of the contact structure **190** described with reference to FIGS. **4A** to **4D** and **6**.

In the example embodiments, the contact structure **190** described with reference to FIGS. **1** to **3D** may be referred to as a first contact structure, and the contact structure **290** described with reference to FIGS. **7** and **8** may be referred to as a second contact structure.

Next, another modified example of the semiconductor device **1** according to an example embodiment is described with reference to FIG. **10**. FIG. **10** is a cross-sectional view illustrating a region taken along line III-III' of FIG. **1**.

Referring to FIGS. **1** and **10**, the modified example may provide a semiconductor device **1c** according to an example embodiment, and the second structure **103** may further include a third contact structure **390** spaced apart from the stack structure **113** without penetrating through the stack structure **113**.

The third contact structure **390** may include the lower contact structure **157ca** penetrating through the capping insulating layer **136**, and the upper contact structure **183c** penetrating through the pattern structure **109** and the upper insulating layer **177**. Here, the lower contact structure **157ca** and the upper contact structure **183c** may be substantially the same as the structures of the lower contact structure **157c** and the upper contact structure **183c** described with reference to FIGS. **1** to **3D**.

In an example embodiment, the third contact structure **390** may be modified like any one of the modified examples of the contact structure **190** described with reference to FIGS. **4A** to **4D** and **6**.

Next, an example of a method of manufacturing a semiconductor device according to an example embodiment is described with reference to FIGS. **11** to **14C**. Among FIGS. **11** to **14C**, FIG. **11** is a cross-sectional view illustrating a method of forming a cross-sectional structure of a first structure **3** in a region taken along lines I-I' and II-II' of FIG. **2**; FIGS. **12A**, **13A** and **14A** are cross-sectional views each illustrating a method of forming a cross-sectional structure of a second structure **103** in the region taken along lines I-I' and II-II' of FIG. **2**; FIGS. **12B**, **13B** and **14B** are cross-sectional views each illustrating a method of forming a cross-sectional structure of the second structure **103** in a region taken along line III-III' of FIG. **1**; and FIGS. **12C**, **13C** and **14C** are cross-sectional views each illustrating a method of forming a cross-sectional structure of a second structure **103a** in a region taken along line IVa-IVa' of FIG. **7**.

Referring to FIG. **11**, the first structure **3** may be formed. The first structure **3** may include: a semiconductor substrate **6**; an isolation region **9s** disposed on the semiconductor substrate **6** and defining a peripheral active region **9a**; a peripheral circuit **12** formed on the semiconductor substrate **6**; a first bonding pads **18** electrically connected to the peripheral circuit **12**; and a first insulating structure **21** disposed on the semiconductor substrate **6**, covering the peripheral circuit **12** and having an upper surface coplanar with an upper surface of the first bonding pads **18**.

The peripheral circuit **12** may include a circuit device **14** such as a transistor including a peripheral gate **14a** and a peripheral source/drain **14b**, and a circuit interconnection **16**

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electrically connected to the circuit device **14**. The circuit interconnection **16** may be electrically connected to the first bonding pads **18**.

Referring to FIGS. **12A**, **12B**, and **12C**, an insulating layer **107** may be formed on a substrate base **105** and a pattern structure **109** may be formed on the insulating layer **107**.

In an example embodiment, the pattern structure **109** may include a polysilicon layer. For example, the pattern structure **109** may include the polysilicon layer having an N-type conductivity.

For another example, the pattern structure **109** may include a polysilicon layer having a P-type conductivity.

For still another example, the pattern structure **109** may include a polysilicon layer including a region having the N-type conductivity and a region having the P-type conductivity. For yet still another example, the pattern structure **109** may include a doped polysilicon layer and a pattern layer disposed on the doped polysilicon layer. The pattern layer may include a first layer, a second layer and a third layer sequentially stacked on one another. Here, the first layer and the third layer may each be a silicon oxide layer, and the second layer may be a silicon layer or a silicon nitride layer.

In another example embodiment, the substrate base **105** and the insulating layer **107** may be omitted, and the pattern structure **109** may include a single crystal silicon layer.

In an example embodiment, an outer insulating layer **111** may be formed on a side surface of the pattern structure **109**.

A preliminary stack structure **115** and a preliminary capping insulating layer **136a** may be formed on the pattern structure **109** and the outer insulating layer **111**.

Forming the preliminary stack structure **115** and the preliminary capping insulating layer **136a** may include the followings: forming a preliminary lower stack structure **116** on the pattern structure **109**; forming a first preliminary upper insulating layer disposed on the pattern structure **109** and the outer insulating layer **111**, covering a portion of the preliminary lower stack structure **116** and having an upper surface coplanar with an upper surface of the preliminary lower stack structure **116**; forming a preliminary upper stack structure **124** on the preliminary lower stack structure **116**; and forming a second preliminary upper insulating layer disposed on the first preliminary upper insulating layer, covering a portion of the preliminary upper stack structure **124** and having an upper surface coplanar with an upper surface of the preliminary upper stack structure **124**. The first and second preliminary upper insulating layers may be formed of the same material as each other, for example, silicon oxide to form the preliminary capping insulating layer **136a**.

Forming the preliminary lower stack structure **116** may include forming second interlayer insulating layers **118** and second preliminary horizontal layers **121** which are alternately stacked in a vertical direction Z, and patterning the second interlayer insulating layers **118** and the second preliminary horizontal layers **121** to form a step shape on at least one side. A lowermost layer and an uppermost layer among the second interlayer insulating layers **118** and the second preliminary horizontal layers **121** may each be one of the second interlayer insulating layers **118**.

Forming the preliminary upper stack structure **124** may include forming first interlayer insulating layers **127** and first preliminary horizontal layers **130** to be alternately stacked in the vertical direction Z, and patterning the first interlayer insulating layers **127** and the first preliminary horizontal layers **130** to form a step shape on at least one side. A lowermost layer and an uppermost layer among the first

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interlayer insulating layers **127** and the first preliminary horizontal layers **130** may each be one of the first interlayer insulating layers **127**.

The first and second interlayer insulating layers **127** and **118** may be formed of a first insulating material layer such as a silicon oxide layer. The first and second preliminary horizontal layers **130** and **121** may be formed of a material different from the first and second interlayer insulating layers **127** and **118**, for example, a second insulating material layer such as a silicon nitride layer. For another example, the first and second preliminary horizontal layers **130** and **121** may be formed of a conductive material layer including at least one of doped polysilicon, metal nitride (e.g., titanium nitride (TiN), tungsten nitride (WN), or tantalum nitride (TaN)), metal-semiconductor compound (e.g., titanium silicide (TiSi), tantalum silicide (TaSi) or nickel silicide (NiSi)), or metal (e.g., tungsten (W)).

The preliminary stack structure **115** may include first preliminary stack structures **114a** and a dummy preliminary stack structure **114b**.

A plurality of memory vertical structures **143c** penetrating through the first preliminary stack structures **114a** may be formed. A dummy vertical structures **143d** penetrating through the second preliminary stack structure **114b** may be formed.

In an example embodiment, the plurality of memory vertical structures **143c** and the dummy vertical structures **143d** may be simultaneously formed. For example, the plurality of memory vertical structures **143c** and the dummy vertical structures **143d** may be formed at the same time and of the same materials.

For another example, the dummy vertical structures **143d** may be formed after the plurality of memory vertical structures **143c** are formed.

Referring to FIGS. **13A**, **13B**, and **13C**, a preliminary capping insulating layer **136b** having an increased upper surface may be formed by forming an additional preliminary capping insulating layer on the preliminary stack structure **115** and the preliminary capping insulating layer **136a**.

Lower contact holes **154c** and separation trenches **154s** which pass through the preliminary capping insulating layer **136b** and the preliminary stack structure **115** may be formed. The lower contact holes **154c** and the separation trenches **154s** may expose the first and second preliminary horizontal layers **130** and **121** of the preliminary stack structure **115**.

An empty space may be formed by etching the first and second preliminary horizontal layers **130** and **121** exposed by the lower contact holes **154c** and the separation trenches **154s**, and first and second horizontal gate layers **131** and **122** may be formed in the respective empty spaces. For example, the second horizontal gate layers **122** may be formed in the empty spaces from which the second preliminary horizontal layers **121** are removed, and the first horizontal gate layers **131** may be formed in the empty spaces from which the first preliminary horizontal layers **130** are removed.

The preliminary lower stack structure **116** may be formed to be a second stacked region **117** including the horizontal gate layers **122** and the interlayer insulating layers **118**, and the preliminary upper stack structure **124** may be formed to be a first stacked region **125** including the horizontal gate layers **131** and the interlayer insulating layers **127**. The stack structure **113** may include the stacked regions **117** and **125**. Here, the horizontal gate layers **122** and interlayer insulating layers **118** of the second stacked region **117** may be referred to as the second horizontal gate layers **122** and the second interlayer insulating layers **118**, respectively, and the horizontal gate layers **131** and interlayer insulating layers **127** of

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the first stacked region **125** may be referred to as the first horizontal gate layers **131** and the first interlayer insulating layers **127**, respectively.

In an example embodiment, a portion of the first preliminary horizontal layers **130** may remain as the horizontal insulating layers **132**.

Lower contact structures **157c** and separation structures **157s** may each be formed in the lower contact holes **154c** and the separation trenches **154s**, respectively.

Forming the lower contact structures **157c** and the separation structures **157s** may include forming insulating spacers **157c1** and **157s1** respectively on sidewalls of the lower contact holes **154c** and separation trenches **154s**, and forming conductive patterns **157c2** and **157s2** respectively filling the lower contact holes **154c** and the separation trenches **154s**. Forming the conductive patterns **157c2** and **157s2** may include forming liner layers **159a1** and **159b1** uniformly covering inner walls of the lower contact holes **154c** and separation trenches **154s** in which the spacers **157c1** and **157s1** are formed, respectively, and forming pillar patterns **159a2** and **159b2** filling the lower contact holes **154c** and the separation trenches **154s**, respectively.

Among the liner layers **159a1** and **159b1**, the liner layer formed in the lower contact holes **154c** may be referred to as the first liner layer **159a1**, and the liner layer formed in the separation trenches **154s** may be referred to as the second liner layer **159b1**. Among the pillar patterns **159a2** and **159b2**, the pillar pattern formed in the lower contact holes **154c** may be referred to as the first pillar pattern **159a2**, and the pillar pattern formed in the separation trenches **154s** may be referred to as the second pillar pattern **159b2**.

Among the spacers **157c1** and **157s1**, the spacer formed on the side surface of the lower contact holes **154c** may be referred to as the lower spacer **157c1**, and the spacer formed on the sidewall of the separation trenches **154s** may be referred to as the separation spacer **157s1**. Among the conductive patterns **157c2** and **157s2**, the pattern formed in the lower contact holes **154c** may be referred to as the lower contact plugs **157c2**, and the conductive patterns formed in the separation trenches **154s** may be referred to as the separation patterns **157s2**.

Referring to FIGS. **14A**, **14B**, and **14C**, a gate contact plugs **162** penetrating through the preliminary capping insulating layer **136b** and electrically connected to the horizontal gate layers **122** and **131** may be formed. The gate contact plugs **162** may be formed in a first stepped region **114s1** in FIG. **1**. A deposition process of depositing the additional capping insulating layer may be performed, and it is possible to form bit line connection vias **164b** passing through the additional capping insulating layer and electrically connected to the memory vertical structures **143c**, gate connection vias **164g** electrically connected to the gate contact plugs **162**, and a contact connection via **164c** electrically connected to the lower contact plug **157c2**. It is possible to form bit lines **168b** electrically connected to the bit line connection vias **164b**, gate interconnections **168g** electrically connected to the gate connection vias **164g**, and a contact interconnection **168c** electrically connected to the contact connection via **164c**.

A capping insulating layer **136** having an increased thickness may be formed by forming an additional capping insulating layer, and it is possible to form an interconnection structure **171** embedded in the capping insulating layer **136** and a second bonding pads **174** on the interconnection structure **171**. The capping insulating layer **136** and the second bonding pads **174** may have upper surfaces coplanar with each other.

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Again referring to FIG. 8 along with FIGS. 1, 2, 3A, and 3B, the structure formed from the capping insulating layer 136 to the second bonding pads 174 may be bonded to the first structure 3 in FIG. 11 by performing a wafer bonding process and using the method described with reference to FIGS. 12A to 14C. Here, the capping insulating layer 136 may be bonded to the first structure 3 while being in contact with the first insulating structure 21, and the second bonding pads 174 may be bonded thereto while being in contact with the first bonding pads 18.

For example, the pattern structure 109 may be exposed by sequentially removing the substrate base 105 and the insulating layer 107, and an upper insulating layer 177 may be formed on the exposed pattern structure 109.

For another example, the substrate base 105 may be removed, and the insulating layer 107 may remain. Here, the remaining insulating layer 107 may be the upper insulating layer 177.

For still another example, when the substrate base 105 and the insulating layer 107 are omitted, and the pattern structure 109 is formed of a single crystal silicon layer, the pattern structure 109 may be formed to have a reduced thickness, and the upper insulating layer 177 may be formed on the pattern structure having the reduced thickness.

The lower contact structure 157c may be exposed while forming an upper contact hole 180 passing through the upper insulating layer 177 and the pattern structure 109, an upper spacer layer covering an inner wall of the upper contact hole 180 may be formed, the upper spacer layer may be anisotropically etched to form an upper spacer 183c1 exposing the lower contact plug 157c2 of the lower contact structure 157c, and an upper contact plug 183c2 filling the contact hole 180 may be formed. Accordingly, an upper contact structure 183 including the upper spacer 183c1 and the upper contact plug 183c2 may be formed. Accordingly, a contact structure 190 including the upper contact structure 183c and the lower contact structure 157c may be formed. A conductive pattern 193 may be formed on the upper contact structure 183c.

In the example embodiments, the contact structure 190 may be formed to have the same shape as any one of cross-sectional structures as illustrated in FIGS. 3D, 4A, 4B, 4C, and 4D depending on a depth at which the lower contact structure 157c extends into the pattern structure 109, a distance between a side surface of the lower contact structure 157c and a side wall of the upper contact hole 180, and a thickness of the upper spacer layer to form the upper spacer 183c1.

Next, an electronic system including a semiconductor device according to an exemplary embodiment is described with reference to FIG. 15. FIG. 15 is a schematic view illustrating the electronic system including the semiconductor device according to an exemplary embodiment.

Referring to FIG. 15, an electronic system 1000 according to an example embodiment may include a semiconductor device 1100 and a controller 1200 electrically connected to the semiconductor device 1100. The electronic system 1000 may be a storage device including one or more semiconductor devices 1100 or an electronic device including the storage device. For example, the electronic system 1000 may be a solid state drive (SSD) device including one or more semiconductor devices 1100, a universal serial bus (USB), a computing system, a medical device, or a communications device.

The semiconductor device 1100 may be a nonvolatile memory device, for example, a semiconductor device according to any one of the example embodiments described

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with reference to FIGS. 1 to 10. The semiconductor device 1100 may include a first structure 1100F and a second structure 1100S disposed on the first structure 1100F.

In an example embodiment, the first structure 1100F may be the first structure 3 of any one of the example embodiments described with reference to FIGS. 1 to 10, and the second structure 1100S may be the second structure 103, 103a, or 103b of any one of the example embodiments described with reference to FIGS. 1 to 10.

The first structure 1100F may be a peripheral circuit structure including a decoder circuit 1110, a page buffer 1120, and a logic circuit 1130. In an example embodiment, the peripheral circuit 12 of the first structure 3 of any one of the example embodiments described with reference to FIGS. 1 to 10 may be a peripheral circuit structure including the decoder circuit 1110, the page buffer 1120, and the logic circuit 1130.

The second structure 1100S may be a memory vertical structure including bit lines BL, a common source line CSL, word lines WL, first and second gate upper lines UL1 and UL2, and first and second gate lower lines LL1 and LL2, and memory cell strings CSTR disposed between the bit lines BL and the common source line CSL.

In an example embodiment, the bit lines BL may be the bit lines 168b of any one of the example embodiments described with reference to FIGS. 1 to 10.

In an example embodiment, the common source line CSL may be a polysilicon layer having an N-type conductivity in at least a portion of the pattern structure 109.

In an example embodiment, the first and second gate lower lines LL1 and LL2, the word lines WL, and the first and second gate upper lines UL1 and UL2 may be the horizontal gate layers 122 and 131 of the first stack portion 113a of any one of the example embodiments described with reference to FIGS. 1 to 10. Therefore, the horizontal gate layers 122 and 131 of the first stack portion 113a may include the first and second gate lower lines LL1 and LL2, the word lines WL, and the first and second gate upper lines UL1 and UL2. At least some of the first and second gate lower lines LL1 and LL2 and the first and second gate upper lines UL1 and UL2 may be selection gate electrodes.

Each of the memory cell strings CSTR of the second structure 1100S may include lower transistors LT1 and LT2 adjacent to the common source line CSL, upper transistors UT1 and UT2 adjacent to the bit line BL, and a plurality of memory cell transistors MCT disposed between the lower transistors LT1 and LT2 and the upper transistors UT1 and UT2. The number of the lower transistors LT1 and LT2 and the number of the upper transistors UT1 and UT2 may be variously modified based on the example embodiments. The plurality of memory cell transistors MCT may each include data storage region capable of storing information (data). For example, the data storage layer 145b of the data storage structure 145 as described with reference to FIG. 5A may include the data storage region.

In the example embodiments, the upper transistors UT1 and UT2 may each include a string select transistor, and the lower transistors LT1 and LT2 may each include a ground select transistor. The gate lower lines LL1 and LL2 may be respective gate electrodes of the lower transistors LT1 and LT2. The word lines WL may be respective gate electrodes of the memory cell transistors MCT, and the gate upper lines UL1 and UL2 may be respective gate electrodes of the upper transistors UT1 and UT2.

In the example embodiments, the lower transistors LT1 and LT2 may include the lower erase control transistor LT1 and the ground select transistor LT2 connected in series with

each other. The upper transistors UT1 and UT2 may include the string select transistor UT1 and the upper erase control transistor UT2 connected in series with each other. At least one of the lower erase control transistor LT1 and the upper erase control transistor UT1 may be used for an erase operation in which data stored in the memory cell transistors MCT is erased using a gate induce drain leakage (GIDL) phenomenon.

The common source line CSL, the first and second gate lower lines LL1 and LL2, the word line WL, and the first and second gate upper lines UL1 and UL2 may each be electrically connected with the decoder circuit 1110 through first interconnections 1115 extending from the inside of the first structure 1100F to the second structure 1100S. The bit line BL may be electrically connected to the page buffer 1120 through second interconnections 1125 extending from the inside of the first structure 1100F to the second structure 1100S.

The decoder circuit 1110 and the page buffer 1120 of the first structure 1100F may perform a control operation on at least one selected memory cell transistor among the plurality of memory cell transistors MCT. The decoder circuit 1110 and the page buffer 1120 may be controlled by the logic circuit 1130. The semiconductor device 1100 may communicate with the controller 1200 through an input/output pad 1101 electrically connected to the logic circuit 1130. The input/output pad 1101 may be electrically connected to the logic circuit 1130 through an input/output connection wiring 1135 extending from the inside of the first structure 1100F to the second structure 1100S.

In an example embodiment, the input/output pad 1101 may be electrically connected to the conductive pattern 193 of any one of the example embodiments described with reference to FIGS. 1 to 10.

In an example embodiment, the input/output connection wirings 1135 may include the contact structures 190, 290, and 390 of any one of the example embodiments described with reference to FIGS. 1 to 10.

The controller 1200 may include a processor 1210, a NAND controller 1220, and a host interface 1230. According to an example embodiment, the electronic system 1000 may include a plurality of semiconductor devices 1100, and in this case, the controller 1200 may control the plurality of semiconductor devices 1100.

The processor 1210 may control an overall operation of the electronic system 1000 including the controller 1200. The processor 1210 may be operated based on predetermined firmware, and may access to the semiconductor device 1100 by controlling the NAND controller 1220. The NAND controller 1220 may include a NAND interface 1221 that processes its communications with the semiconductor device 1100. Through the NAND interface 1221, it is possible to transmit a control command for controlling the semiconductor device 1100, data to be written to the memory cell transistor MCT of the semiconductor device 1100, data to be read from the memory cell transistor MCT of the semiconductor device 1100, etc. The host interface 1230 may provide communications between the electronic system 1000 and an external host. When the control command is received from the external host through the host interface 1230, the processor 1210 may control the semiconductor device 1100 in response to the control command.

The electronic system including a semiconductor device according to another exemplary embodiment is described with reference to FIG. 16. FIG. 16 is a schematic perspective

view illustrating the electronic system including the semiconductor device according to another example embodiment.

Referring to FIG. 16, an electronic system 2000 according to an example embodiment may include a main substrate 2001, a controller 2002 mounted on the main substrate 2001, one or more semiconductor packages 2003, and a dynamic random access memory (DRAM) 2004. The semiconductor package 2003 and the DRAM 2004 may be connected to the controller 2002 by a wiring pattern 2005 formed on the main substrate 2001.

The main substrate 2001 may include a connector 2006 including a plurality of pins coupled to an external host. The number and arrangement of the plurality of pins on the connector 2006 may depend on a communications interface between the electronic system 2000 and the external host. In the example embodiments, the electronic system 2000 may communicate with the external host based on any one of the interfaces such as a universal serial bus (USB), a peripheral component interconnect (PCI)-express, a serial advanced technology attachment (SATA), and an M-physostigmine (phy) for a universal flash storage (UFS). In the example embodiments, the electronic system 2000 may be operated by power supplied from the external host through the connector 2006. The electronic system 2000 may further include a power management integrated circuit (PMIC) that distributes the power supplied from the external host to the controller 2002 and the semiconductor package 2003.

The controller 2002 may write data to the semiconductor package 2003 or read data from the semiconductor package 2003, and may improve an operation speed of the electronic system 2000.

The DRAM 2004 may be a buffer memory to mitigate a difference in speed between the semiconductor package 2003, which is a data storage space, and the external host. The DRAM 2004 included in the electronic system 2000 may also be operated as a type of a cache memory, and may provide a space for temporarily storing data during an operation of controlling the semiconductor package 2003. When the DRAM 2004 is included in the electronic system 2000, the controller 2002 may further include a DRAM controller controlling the DRAM 2004 in addition to the NAND controller controlling the semiconductor package 2003.

The semiconductor package 2003 may include first and second semiconductor packages 2003a and 2003b spaced apart from each other. Each of the first and second semiconductor packages 2003a and 2003b may be a semiconductor package including a plurality of semiconductor chips 2200. Each of the first and second semiconductor packages 2003a and 2003b may include: a package substrate 2100; the semiconductor chips 2200 disposed on the package substrate 2100; an adhesive layer 2300 disposed on a lower surface of each of the semiconductor chips 2200; a connection structure 2400 electrically connecting the semiconductor chips 2200 and the package substrate 2100 to each other; and a molding layer 2500 disposed on the package substrate 2100 and covering the semiconductor chips 2200 and the connection structure 2400.

The package substrate 2100 may be a printed circuit board including a package upper pad 2130. Each of the semiconductor chips 2200 may include an input/output pad 2210. The input/output pad 2210 may correspond to the input/output pad 1101 of FIG. 15. Each of the semiconductor chips 2200 may include stack structures 3210 and memory vertical structures 3220. Each of the semiconductor chips 2200 may include the semiconductor device of any one of the example

embodiments described with reference to FIGS. 1 to 10. The stack structure 3210 may be the stack structure 113 of any one of the example embodiments described with reference to FIGS. 1 to 10. The memory vertical structures 3220 may be the memory vertical structures 143c or 143c' of any one of the example embodiments described with reference to FIGS. 1 to 10.

In the example embodiments, the connection structure 2400 may be a bonding wire electrically connecting the input/output pad 2210 and the upper pad 2130 of the package to each other. Accordingly, the semiconductor chips 2200 of each of the first and second semiconductor packages 2003a and 2003b may be electrically connected to each other by a bonding wire method, and may each be electrically connected to the package upper pad 2130 of the package substrate 2100. According to an example embodiment, the semiconductor chips 2200 of each of the first and second semiconductor packages 2003a and 2003b may also be electrically connected to each other by a connection structure including a through electrode (e.g., through silicon via (TSV)), instead of the bonding wire type connection structure 2400.

In the example embodiments, the controller 2002 and the semiconductor chip 2200 may be included in one package. In an example embodiment, the controller 2002 and the semiconductor chips 2200 may be mounted on a separate interposer substrate different from the main substrate 2001, and the controller 2002 and the semiconductor chips 2200 may be connected to each other by a wiring formed on the interposer substrate.

FIG. 17 is a schematic cross-sectional view illustrating a semiconductor package according to an exemplary embodiment. FIG. 17 illustrates an example embodiment of the semiconductor package 2003 of FIG. 16, and conceptually illustrates a region cut along a cutting line V-V' of the semiconductor package 2003 of FIG. 16.

Referring to FIG. 17, each of the semiconductor chips 2200a of the semiconductor package 2003A may include a semiconductor substrate 4010 and a first structure 4100 disposed on the semiconductor substrate 4010, and a second structure 4200 disposed on the first structure 4100 and bonded to the first structure 4100 by the wafer bonding method.

In an example embodiment, the first structure 4100 may be the first structure 3 of any one of the example embodiments described with reference to FIGS. 1 to 10 and/or the first structure 1100F described with reference to FIG. 15, and the second structure 4200 may be the second structure 103, 103a, or 103b of any one of the example embodiments described with reference to FIGS. 1 to 10 and/or the second structure 1100S described with reference to FIG. 15.

The first structure 4100 may include a peripheral circuit region including a peripheral wiring 4110 and first bonded structures 4150. The second structure 4200 may include: a common source line 4205; a gate stack structure 4210 disposed between the common source line 4205 and the first structure 4100; memory vertical structures 4220 and separation structures 4230 which penetrate through the gate stack structure 4210; and second bonded structures 4250 electrically connected to each of the word lines (e.g., the word lines WL in FIG. 15) of the gate stack structure 4210 and the memory vertical structure 4220. For example, the second bonded structures 4250 may be electrically connected to each of the memory vertical structures 4220 and the word lines (e.g., the word lines WL in FIG. 15) through bit lines 4240 electrically connected to the memory vertical structures 4220 and the gate interconnections (e.g., the gate

interconnections 168g in FIG. 3B) electrically connected to the word lines (e.g., the word lines WL in FIG. 15). The first bonded structures 4150 of the first structure 4100 and the second bonded structures 4250 of the second structure 4200 may be bonded to each other while being in contact with each other. A portion in which the first bonded structures 4150 and the second bonded structures 4250 are bonded to each other may be formed of, for example, copper (Cu).

In an example embodiment, the gate stack structure 4210 may be the stack structure 113 of any one of the example embodiments described with reference to FIGS. 1 to 10.

In an example embodiment, the memory vertical structures 4220 may be the memory vertical structures 143c or 143c' of any one of the example embodiments described with reference to FIGS. 1 to 10.

In an example embodiment, the separation structures 4230 may be the separation structures 157s of any one of the example embodiments described with reference to FIGS. 1 to 10.

In an example embodiment, the first bonded structures 4150 may include the first bonding pads 18 of any one of the example embodiments described with reference to FIGS. 1 to 10, and the second bonded structures 4250 may include the second bonding pads 174 of any one of the example embodiments described with reference to FIGS. 1 to 10.

Each of the semiconductor chips 2200a may further include the input/output pad 2210. The input/output pad 2210 may be electrically connected to the conductive pattern 193 of any one of the example embodiments described with reference to FIGS. 1 to 10. The semiconductor chips 2200a may be electrically connected to each other by the bonding wire type connection structure 2400. However, in the example embodiments, semiconductor chips in one semiconductor package, such as the semiconductor chips 2200a, may be electrically connected to each other by the connection structure including the through electrode (through silicon via, TSV).

As set forth above, the example embodiments may provide the semiconductor device having improved integration and reliability and the data storage system including the same. For example, it is possible to penetrate through the horizontal conductive layers formed simultaneously as the word lines, and to provide the contact structure formed simultaneously as the separation structures. Accordingly, it is thus possible to form the contact structure more stably and minimize the space occupied by the contact structure, thereby improving a degree of integration of the semiconductor device. As a result, the semiconductor device may have the improved integration density and reliability.

The various and beneficial advantages and effects of the present inventive concept are not limited to the above description, and more easily understood in the process of explaining the specific example embodiments.

While example embodiments have been illustrated and described above, it will be apparent to those skilled in the art that modifications and variations could be made without departing from the scope of the present inventive concept as defined by the appended claims.

What is claimed is:

1. A semiconductor device comprising:
 - a first structure including a peripheral circuit; and
 - a second structure disposed on the first structure and bonded to the first structure,
 wherein the second structure includes:
 - a pattern structure including a polysilicon layer;
 - an upper insulating layer disposed on the pattern structure;

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a stack structure disposed between the first structure and the pattern structure and including a first stack portion and a second stack portion spaced apart from each other in a horizontal direction, the first and second stack portions respectively including horizontal conductive layers and interlayer insulating layers alternately stacked in a vertical direction; separation structures penetrating through the stack structure and separating the stack structure; memory vertical structures penetrating through the first stack portion of the stack structure; bit lines below the memory vertical structures and electrically connected to the memory vertical structures, and

a contact structure penetrating through the second stack portion of the stack structure, the pattern structure, and the upper insulating layer,

wherein the contact structure includes:

a lower contact plug penetrating through at least the second stack portion of the stack structure and spaced apart from the horizontal conductive layers of the second stack portion; and

an upper contact plug in contact with the lower contact plug and extending upwardly to penetrate through the pattern structure and the upper insulating layer, wherein the polysilicon layer of the pattern structure is spaced apart from the lower contact plug and the upper contact plug and is in contact with the separation structures, and

wherein the bit lines are not electrically connected to the contact structure and the separation structures.

2. The semiconductor device of claim 1, wherein the upper contact plug is in contact with an upper surface of the lower contact plug and a side surface of the lower contact plug, adjacent to the upper surface thereof.

3. The semiconductor device of claim 2, wherein the lower contact plug further includes a portion extending into the pattern structure from a portion penetrating through the second stack portion of the stack structure, and

wherein the upper surface of the lower contact plug is disposed at a height level between a lower surface of the pattern structure and an upper surface of the pattern structure.

4. The semiconductor device of claim 3, wherein a height difference between the upper surface of the lower contact plug and the lower surface of the pattern structure is smaller than a height difference between the upper surface of the lower contact plug and the upper surface of the pattern structure.

5. The semiconductor device of claim 1, wherein the contact structure further includes an insulating lower spacer covering a side surface of the lower contact plug, and an insulating upper spacer covering a side surface of the upper contact plug, and

wherein a thickness of the insulating upper spacer in the horizontal direction is greater than a thickness of the insulating lower spacer in the horizontal direction.

6. The semiconductor device of claim 1, wherein the contact structure further includes an insulating lower spacer covering a side surface of the lower contact plug,

wherein each of the separation structures includes a conductive separation pattern and an insulating separation spacer covering a side surface of the conductive separation pattern,

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wherein the lower contact plug includes a first liner layer and a first pillar pattern,

wherein the conductive separation pattern includes a second liner layer formed of the same material as the first liner layer and a second pillar pattern formed of the same material as the first pillar pattern,

wherein the first liner layer covers at least a side surface of the first pillar pattern, and

wherein the second liner layer covers a side surface of the second pillar pattern and an upper surface of the second pillar pattern.

7. The semiconductor device of claim 6, wherein the first liner layer includes a portion interposed between the first pillar pattern and the upper contact plug, and

wherein the first pillar pattern is spaced apart from the upper contact plug.

8. The semiconductor device of claim 6, wherein the first pillar pattern is in contact with the upper contact plug.

9. The semiconductor device of claim 1, wherein the first structure includes first bonding pads, and wherein the second structure further includes:

gate contact plugs disposed under the stack structure and electrically connected to word lines and selection gate electrodes of the horizontal conductive layers of the first stack portion;

bit lines disposed between the memory vertical structures and the first structure, and electrically connected to the memory vertical structures;

gate interconnections disposed between the gate contact plugs and the first structure, and electrically connected to the gate contact plugs;

a contact interconnection disposed between the contact structure and the first structure, and electrically connected to the lower contact plug;

second bonding pads in contact with and bonded to the first bonding pads; and

an interconnection structure electrically connecting the second bonding pads with the bit lines, the gate interconnections, and the contact interconnection.

10. The semiconductor device of claim 9, wherein the first stack portion includes a first stacked region and a second stacked region disposed on the first stacked region,

wherein the first stacked region includes a plurality of first horizontal conductive layers of the horizontal conductive layers,

wherein the second stacked region includes a plurality of second horizontal conductive layers of the horizontal conductive layers, and

wherein a side surface of at least one of the memory vertical structures has a bent portion between the plurality of first horizontal conductive layers and the plurality of second horizontal conductive layers.

11. The semiconductor device of claim 1, wherein one of the separation structures is disposed between the first stack portion and the second stack portion.

12. The semiconductor device of claim 1, wherein the second structure further includes a capping insulating layer,

wherein the stack structure includes a first stack structure including the first stack portion and a dummy stack structure including the second stack portion,

wherein the memory vertical structures do not penetrate through the dummy stack structure, and

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wherein at least a portion of the capping insulating layer is disposed between the first stack structure and the dummy stack structure.

13. The semiconductor device of claim 1,

wherein the pattern structure has a lower surface facing the first structure and an upper surface opposing the lower surface,

wherein each of the separation structures includes a conductive separation pattern and an insulating separation spacer covering a side surface of the conductive separation pattern,

wherein the conductive separation pattern of at least one of the separation structures further includes a portion extending into the pattern structure from a portion penetrating through the stack structure,

wherein the conductive separation pattern has an upper surface disposed at a height level between an upper surface of the pattern structure and a lower surface of the pattern structure, and

wherein a height difference between the upper surface of the conductive separation pattern and the lower surface of the pattern structure is greater than a height difference between an upper surface of the lower contact plug and a lower surface of the pattern structure.

14. The semiconductor device of claim 1,

wherein the second structure further includes:

bit lines disposed between the stack structure and the first structure; and

dummy vertical structures penetrating through the second stack portion, and

wherein the bit lines are electrically connected to the memory vertical structures, and are not electrically connected to the dummy vertical structures.

15. A semiconductor device comprising:

a first structure including a peripheral circuit; and

a second structure disposed on the first structure and bonded to the first structure,

wherein the second structure includes:

a pattern structure including a polysilicon layer; an upper insulating layer disposed on the pattern structure;

word lines disposed between the pattern structure and the first structure, stacked in a vertical direction and spaced apart from each other;

dummy horizontal conductive layers disposed between the pattern structure and the first structure, stacked in the vertical direction and spaced apart from each other;

memory vertical structures penetrating through the word lines in the vertical direction and in contact with the pattern structure;

bit lines below the memory vertical structures and electrically connected to the memory vertical structures;

separation structures penetrating through the word lines in the vertical direction and in contact with the pattern structure; and

a contact structure penetrating through the dummy horizontal conductive layers, the pattern structure, and the upper insulating layer in the vertical direction,

wherein the dummy horizontal conductive layers are electrically isolated from the contact structure,

wherein the contact structure includes a lower contact structure and an upper contact structure,

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wherein the lower contact structure includes a lower contact plug penetrating through at least the dummy horizontal conductive layers,

wherein the upper contact structure includes an upper contact plug in contact with the lower contact plug and extending upwardly to penetrate through the pattern structure,

wherein the lower contact plug is spaced apart from the dummy horizontal conductive layers,

wherein the polysilicon layer of the pattern structure is spaced apart from the lower contact plug and the upper contact plug and is in contact with the separation structures, and

wherein the bit lines are not electrically connected to the contact structure and the separation structures.

16. The semiconductor device of claim 15,

wherein each of the separation structures includes a conductive separation pattern and an insulating separation spacer covering a side surface of the conductive separation pattern,

wherein the lower contact structure further includes an insulating lower spacer surrounding at least a side surface of the lower contact plug,

wherein the upper contact structure further includes an insulating upper spacer surrounding a side surface of the upper contact plug,

wherein the upper contact plug is in contact with an upper surface of the lower contact plug and the side surface of the lower contact plug, adjacent to the upper surface thereof, and

wherein each of the conductive separation pattern and the lower contact plug includes a pillar pattern and a liner layer covering at least a side surface of the pillar pattern.

17. The semiconductor device of claim 15,

wherein the second structure further includes dummy vertical structures penetrating through the dummy horizontal conductive layers and spaced apart from each other, and

wherein the dummy vertical structures are arranged to surround a portion of the contact structure penetrating through the dummy horizontal conductive layers.

18. A data storage system comprising:

a semiconductor device; and

a controller electrically connected to the semiconductor device,

wherein the semiconductor device includes:

a first structure including a peripheral circuit; and

a second structure disposed on the first structure and bonded to the first structure,

wherein the second structure includes:

a pattern structure including a polysilicon layer;

an upper insulating layer disposed on the pattern structure;

word lines disposed between the pattern structure and the first structure, and

stacked in a vertical direction;

dummy horizontal conductive layers disposed between the pattern structure and

the first structure, and stacked in the vertical direction;

memory vertical structures penetrating through the word lines in the vertical direction and in contact with the pattern structure;

separation structures penetrating through the word lines in the vertical direction and in contact with the pattern structure; and

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a contact structure penetrating through the dummy horizontal conductive layers, the pattern structure, and the upper insulating layer in the vertical direction,

wherein the dummy horizontal conductive layers are electrically isolated from the contact structure,

wherein the contact structure includes a lower contact structure and an upper contact structure,

wherein the lower contact structure includes a lower contact plug penetrating through at least the dummy horizontal conductive layers,

wherein the upper contact structure includes an upper contact plug in contact with the lower contact plug and extending upwardly to penetrate through the pattern structure,

wherein the lower contact plug is spaced apart from the dummy horizontal conductive layers, and

wherein the polysilicon layer of the pattern structure is spaced apart from the lower contact plug and the upper contact plug and is in contact with the separation structures.

19. The data storage system of claim 18,

wherein the second structure further includes dummy vertical structures penetrating through the dummy horizontal conductive layers and spaced apart from each other,

wherein each of the separation structures includes a conductive separation pattern and an insulating separation spacer covering a side surface of the conductive separation pattern,

wherein the lower contact structure further includes an insulating lower spacer surrounding at least a side surface of the lower contact plug,

wherein the upper contact structure further includes an insulating upper spacer surrounding a side surface of the upper contact plug,

wherein the upper contact plug is in contact with an upper surface of the lower contact plug and the side surface of the lower contact plug, adjacent to the upper surface thereof, and

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wherein each of the conductive separation pattern and the lower contact plug includes a pillar pattern and a liner layer covering at least a side surface of the pillar pattern.

20. The data storage system of claim 18,

wherein the lower contact structure further includes an insulating lower spacer surrounding at least a side surface of the lower contact plug,

wherein the upper contact structure further includes an insulating upper spacer surrounding a side surface of the upper contact plug,

wherein the first structure includes first bonding pads, wherein the second structure further includes:

an input/output pad disposed on the upper insulating layer and electrically connected to the contact structure;

gate contact plugs disposed between the word lines and the first structure and electrically connected to the word lines;

bit lines disposed between the memory vertical structures and the first structure, and electrically connected to the memory vertical structures;

gate interconnections disposed between the gate contact plugs and the first structure, and electrically connected to the gate contact plugs;

a contact interconnection disposed between the contact structure and the first structure, and electrically connected to the lower contact plug;

second bonding pads in contact with and bonded to the first bonding pads; and

an interconnection structure electrically connecting the second bonding pads with the bit lines, the gate interconnections, and the contact interconnection,

wherein each of the memory vertical structures includes the a data storage layer, and

wherein the semiconductor device is electrically connected to the controller through the input/output pad.

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